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SURVEY

Integrated Sensing and Communications: A Research Roadmap

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ABSTRACT Integrated sensing and communications (ISAC) is emerging as a major architectural direction for next-generation wireless systems. By jointly designing environmental sensing and wireless data transmission within a unified framework, ISAC offers the potential for simultaneous gains in spectral efficiency, hardware reuse, energy efficiency, and situational awareness. This paper provides a comprehensive research roadmap for ISAC, tracing its evolution from radar–communication coexistence to dual-function radar–communication architectures, networked sensing infrastructures, and perceptive sixth-generation (6G) wireless networks. The survey is organized around five pillars: information-theoretic foundations, physical-layer design, networked operation, enabling platform technologies, and application domains. It further examines emerging regimes and deployment extensions that challenge conventional assumptions, including near-field operation, terahertz systems, AI-native transceivers, reconfigurable and digital-twin-assisted environments, low-power and backscatter-based sensing, and non-terrestrial ISAC. Across these topics, the paper identifies open gaps, formulates a unified set of grand research challenges, reviews representative prototype and measurement-oriented studies, and outlines both near-term deployment paths and longer-horizon transformative directions. The goal is to provide a coherent reference and roadmap for the continued development of ISAC as a foundation for future wireless, sensing, and cyber-physical systems.

INDEX TERMS 6G, ISAC, dual-function radar–communication, networked ISAC, cell-free ISAC, near-field ISAC, terahertz communications, reconfigurable intelligent surfaces, AI-native transceivers, digital twins, non-terrestrial networks, energy efficiency, physical-layer security.

I. INTRODUCTION

A. MOTIVATION AND SIGNIFICANCE

The history of wireless technology is punctuated by moments of conceptual re-imagination. The transition from analog to digital modulation, the rise of multiple-input multiple-output (MIMO) processing, and the adoption of orthogonal frequency-division multiplexing (OFDM) each expanded what wireless systems could accomplish. Integrated sensing and communications (ISAC) represents the next such shift. For most of the twentieth century, radar and communication evolved on separate tracks [1]. Radar systems were optimized for target detection, ranging, and tracking, whereas communication systems were optimized for throughput, reliability, and spectral efficiency. Although the two communities shared mathematical tools from signal processing and information theory, their systems shared little else—neither spectrum, nor hardware, nor design philosophy.

This separation was neither accidental nor irrational. The requirements of the two functions appeared fundamentally mismatched. Radar favored deterministic probing signals, wideband ranging resolution, and sensing-oriented beam patterns, while communication favored randomized codebooks, multiplexing efficiency, and user-centric resource allocation. Under these assumptions, joint design appeared to entail unavoidable performance sacrifice. Modern ISAC changes that perspective by treating sensing and communication as co-designed functions within a common transceiver and

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signaling architecture [2], [3], [4]. Its core premise is that sensing and communication need not be treated as isolated functionalities that merely coexist in the same spectrum. Instead, they can be jointly designed so that the same waveform, hardware platform, and spatial processing architecture support both environmental inference and data transmission. In that sense, ISAC is significant not only because it improves spectral reuse, but because it redefines what a wireless infrastructure is expected to do: not simply connect devices, but also perceive the physical world.

B. WHY ISAC NOW

The renewed urgency of ISAC is driven by three converging forces. The first is spectrum scarcity. Radar and communication systems have expanded into increasingly overlapping frequency bands, including S-band, C-band, millimeter-wave (mmWave), and, increasingly, terahertz (THz) ranges. As a result, mutual interference can no longer be treated as an exceptional edge case; it has become a structural constraint on future wireless design. Joint operations are therefore increasingly attractive not merely as an efficiency improvement, but also as a practical response to growing competition for RF resources. This urgency is reinforced by the broader 6G vision, in which communication, sensing, and environmental awareness are increasingly treated as integrated capabilities rather than separate services [5].

The second force is hardware convergence. Advances in silicon RF integrated circuits, phased arrays, massive-MIMO platforms, and software-defined radio have made it both economically attractive and technically feasible to implement sensing and communication on shared hardware. The same aperture can now support directional communication and target illumination, while the same baseband stack can increasingly accommodate estimation, beam control, and joint processing. This removes one of the historical barriers that kept the two functions separate. More broadly, ISAC should be understood as part of the longer transition from communication-only mobile networks toward wireless infrastructures that are increasingly programmable, context-aware, and tightly integrated with sensing and computation [6].

The third force is application demand. Many of the most important 6G use cases require simultaneous communication and sensing rather than independent operation. Autonomous mobility, industrial automation, smart-city infrastructure, immersive reality, and environmental monitoring all demand not only reliable connectivity, but also fine-grained awareness of position, motion, blockage, occupancy, and scene dynamics [7], [8]. In such applications, communication without perception is insufficient, while sensing without communication is operationally incomplete. ISAC therefore emerges not as a convenience, but as a natural architectural response to the joint requirements of next-generation wireless systems.

C. EMERGING REGIMES AND NEW FOUNDATIONS

Much of the existing ISAC literature has implicitly inherited three assumptions from conventional wireless design: far-field propagation, narrow relative bandwidth, and modular transceiver architecture. Under these assumptions, wavefronts are approximately planar, beamforming is approximately frequency-flat, and the transceiver can be decomposed into relatively independent blocks such as channel estimation, equalization, decoding, and sensing. These assumptions have been analytically convenient, but they are becoming increasingly fragile in the face of 6G-scale hardware, bandwidth, and intelligence requirements. Figure 1 summarizes this motivation by linking the historical evolution of ISAC, the current drivers of adoption, and the emerging regimes that motivate the roadmap.

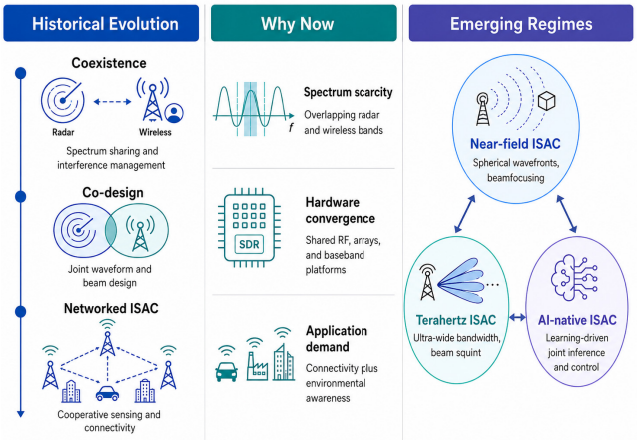


FIGURE 1. Historical evolution, current drivers, and emerging regimes that motivate the ISAC research roadmap.

This paper argues that the next phase of ISAC research is being shaped by emerging regimes that break these assumptions. The first is near-field ISAC. As apertures become larger and frequencies higher, many users and targets move into the Fresnel region, where spherical wavefronts invalidate far-field angle-only models [9], [10]. This breaks the separation between range and angle, but also creates a new opportunity: beamfocusing can encode range into the spatial phase profile, enabling richer localization and environmental sensing. The second is THz ISAC. THz bands offer enormous contiguous bandwidth, enabling ultra-high-range resolution and micro-Doppler sensitivity [11], [12]. At the same time, THz operation introduces molecular-absorption-induced transmission windows, strong propagation fragility, and beam squint due to the large relative bandwidth. As a result, frequency and spatial processing cannot be optimized independently. The third is AI-native ISAC. Machine learning has been used to assist beamforming, resource allocation, and channel inference, but the emerging vision is more radical: end-to-end learned transceivers that map raw observations directly to communication and sensing outputs [13], [14]. Such architectures may bypass hand-crafted modular tradeoffs, but they also raise new questions of robustness, certification, and proximity to information-theoretic limits.

These regimes are not peripheral edge cases. They represent the frontier at which many of the largest future gains in ISAC are likely to be realized. They also interact. Near-field effects become particularly pronounced at high frequencies because of large aperture-to-wavelength ratios, while AI-native processing may become essential for handling the complexity of THz-wideband beam control and model mismatch. A modern treatment of ISAC must therefore go beyond conventional far-field, narrowband, and modular assumptions, and instead develop new foundations suited to these emerging operating regimes.

D. CONTRIBUTIONS, SCOPE, AND PAPER ORGANIZATION

This paper has four primary goals. First, it provides a structured and forward-looking survey of the ISAC literature, organized not merely by topic but by the logical development of the field from conceptual foundations to physical-layer design, networked systems, applications, and future directions. Second, it identifies the major unresolved research gaps that currently prevent ISAC from maturing into a fully deployable 6G capability. Third, it formulates a unified set of grand research challenges that connect theory, hardware, architecture, security, and cross-layer intelligence. Fourth, it outlines a set of transformative research directions that may redefine the field over the longer term.

The scope of the paper is terrestrial RF ISAC spanning sub-6 GHz, mmWave, and lower-THz systems. The emphasis is on wireless infrastructures and architectures in which sensing and communication are jointly designed at the signal, transceiver, network, or systems level. Several adjacent domains are left outside the main scope, including optical ISAC, underwater acoustic sensing and communications, detailed circuit-level implementation below the architecture level, and broader regulatory or policy analysis. Low-power and low-power wide-area network (LPWAN)-style ISAC systems are discussed only briefly as energy-centric boundary cases because they expose a distinct regime in which sensing is opportunistic, coarse-grained, and constrained by multi-year battery-life requirements.

The remainder of the paper is organized as follows. Section II reviews the historical, conceptual, and information-theoretic foundations of ISAC and situates this paper relative to the existing survey literature. Section III examines physical-layer design, including waveform design, beamforming, and channel modeling. Section IV examines networked and distributed ISAC, including cell-free architectures, fronthaul, synchronization, interference, and system-level scaling. Section V develops the emerging-regime and enabling-technology perspective, including near-field operation, reconfigurable intelligent surface (RIS)-enabled ISAC, mmWave/THz systems, AI-native transceivers, energy-efficient design, digital twins, and low-power ISAC. Section VI reviews applications, prototypes, and the standardization outlook. Section VII synthesizes the grand research challenges that define the field's main unresolved bottlenecks, while Section VIII discusses transformative directions that may reshape ISAC over the next decade. Section IX concludes the paper with an integrated outlook.

II. FOUNDATIONS AND RELATED WORK

The rise of ISAC has produced a broad and fast-moving literature spanning information theory, waveform and beamforming design, networked sensing architectures, RIS-enabled systems, security, machine learning, and standardization. As a result, a modern survey must do more than summarize isolated technical threads. It must clarify how the field evolved conceptually, how existing surveys differ in scope and emphasis, and where the main unresolved gaps now lie. This section serves two purposes: it situates the present paper within the existing ISAC survey landscape and establishes the historical and conceptual foundation for the technical sections that follow.

A. FROM COEXISTENCE TO CO-DESIGN: HISTORICAL EVOLUTION

The intellectual roots of ISAC can be traced to the early 2000s, when researchers first confronted growing spectrum overlap between civilian radar systems and emerging broadband wireless deployments. In that period, the dominant question was defensive rather than integrative: how can radar and communications operate in adjacent or shared spectrum without causing unacceptable mutual interference? The resulting research agenda focused on coexistence mechanisms, including spatial separation, null steering, interference avoidance, and power control. This early phase was shaped by spectrum-engineering concerns more than by unified transceiver design, and is best understood as a coexistence problem rather than as true functional integration [15].

The conceptual turning point occurred roughly during the 2009–2012 period, with the emergence of dual-function radar-communication (DFRC) thinking. The key shift was not merely technical, but conceptual. Instead of asking how two separate systems could peacefully coexist, DFRC asked whether a single transmitted waveform or transceiver architecture could simultaneously support both sensing and communication. This reframing transformed the problem from one of coordination to one of co-design. Once this possibility was recognized, the field began to move away from spectrum sharing as its primary narrative and toward the deeper question of how sensing and communication could be jointly optimized within a common physical-layer architecture. This transition is reflected clearly in later tutorials and survey treatments that identify DFRC as the conceptual bridge from coexistence to unified design [2], [4].

That transition initiated the second major phase of the field, in which co-design became the dominant research paradigm. During this period, work on waveform design, beamforming, and information-theoretic tradeoffs began to establish that communication and sensing need not always compete through rigid resource partitioning. Instead, carefully designed waveforms and beamformers could often embed sensing functionality directly into

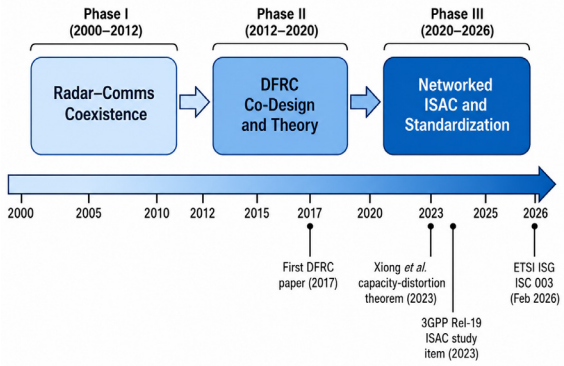


FIGURE 2. Three-phase historical evolution of ISAC, from radar–communications coexistence through DFRC co-design to networked ISAC and standardization. The figure highlights the shift from interference management between separate systems to unified transceiver design and finally to infrastructure-scale integration and standards-facing development.

communication-bearing signals. This insight led to a substantial expansion of the literature, including more systematic optimization frameworks, early prototypes, and the first rigorous formulations of sensing–communication tradeoffs [4], [5]. In hindsight, this phase established the theoretical and architectural core of what is now widely understood as ISAC.

The current phase extends beyond single-node DFRC toward networked ISAC and standardization. Once dense cellular infrastructures, distributed access points, cooperative sensing geometries, and 5G/6G signaling structures were introduced into the discussion, the field widened again. ISAC became not only a waveform or transceiver problem, but also a systems problem involving synchronization, fronthaul, interference coordination, deployment compatibility, and standardization pathways. This is also the phase in which institutional momentum became visible. The emergence of exploratory 3GPP study items, followed by broader 6G-facing study activities and the release of ETSI ISAC reference material, signaled that the field was no longer confined to academic curiosity. The trajectory from coexistence to co-design has therefore continued into a third stage in which networked sensing, infrastructure support, and standards-facing architecture are becoming central.

Figure 2 summarizes this evolution in three phases: an initial coexistence era, a subsequent DFRC co-design era, and the present movement toward networked ISAC and standardization. This three-phase view is useful not because the field changed abruptly at exact dates, but because it captures the dominant emphasis of each period. The first phase treated radar and communications as neighbors; the second treated them as jointly designed functions; the third treats them as components of a broader networked wireless infrastructure.

This historical trajectory also clarifies why current ISAC research often appears to mix very different design philosophies. Coexistence-oriented methods remain relevant in legacy and heterogeneous spectrum settings. DFRC remains the dominant paradigm for single-node waveform and beamforming design. Networked ISAC, in turn, introduces concerns that were largely absent from early DFRC work, including distributed sensing geometry, cooperative processing, and standardization constraints. Rather than viewing these as competing definitions of ISAC, it is more accurate to interpret them as successive layers of the field's maturation. This perspective provides a useful foundation for the related-work discussion that follows and for the broader roadmap developed in the remainder of the paper.

B. THE LANDSCAPE OF EXISTING ISAC SURVEYS

The rapid maturation of ISAC has generated a correspondingly broad survey literature, but that literature remains fragmented in scope. Some surveys emphasize taxonomy and conceptual classification, others focus on physical-layer DFRC design, while more recent works cover vehicular ISAC, RIS-assisted systems, security and privacy, prototype validation, or selected standardization developments. Table 1 compares the present paper with representative existing surveys across six technical dimensions: information-theoretic treatment, physical-layer design, RIS and platform technologies, networked ISAC, security and privacy, and standardization/prototype coverage.

Literature-selection protocol and evidence boundary. To make this comparison reproducible, we used a structured literature-selection process. We considered peer-reviewed journal and conference papers, tutorial and survey articles, standards-facing documents, and selected preprints available at the time of revision, using May 2026 as the literature cutoff. The main sources included IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, arXiv, 3GPP, ETSI, and ITU-R repositories. Preprints were included only when they addressed recent topics not yet covered in archival surveys, and standards documents were used to characterize study-stage or standards-facing activity rather than to imply finalized deployment commitments. Representative search terms included "integrated sensing and communications," "ISAC," "dual-function radar communication," "DFRC," "networked ISAC," "cell-free ISAC," "RIS-assisted ISAC," "near-field ISAC," "THz ISAC," "semantic communication," "sensing-assisted communication," "NTN ISAC," and "ISAC standardization." We included works that provided foundational theory, physical-layer design, networked architectures, enabling platforms, security/privacy analysis, prototype validation, or standards-facing discussion. Works focused primarily on optical ISAC, underwater acoustic systems, detailed circuit-level implementation, or regulatory-policy analysis were excluded except when needed for boundary context. Throughout the paper, experimentally supported results, literature-based synthesis, illustrative numerical examples, and long-horizon speculation are distinguished explicitly in the text or figure captions.

Early surveys played an important role in establishing the field's vocabulary. The earliest taxonomic works clarified the

TABLE 1. Comparison of representative ISAC survey and roadmap papers selected according to the literature-selection protocol described in Section II-B.

Survey	Focus	Info. Theory	PHY Design	RIS/Platform	Net. ISAC	Sec./Privacy	Std./Proto.	Limitation
Liu et al. [2]	Broad foundational ISAC/DFRC tutorial and taxonomy.	○	✓	–	○	–	–	Written before recent standards-facing ISAC, NTN-ISAC, digital-twin, and long-horizon roadmap discussions became central.
Zhang et al. [3]	Point-to-point DFRC signal-processing design.	–	✓	–	–	–	–	Focused mainly on co-located or point-to-point DFRC rather than networked, standards-oriented, and emerging-platform ISAC.
A. Liu et al. [16]	Fundamental limits and information-theoretic treatment.	✓	○	–	○	–	–	Strong on theory, but limited coverage of RIS/platform technologies, prototypes, standardization, and deployment-oriented roadmap issues.
Wei et al. [17]	Sensing–communication–computing integration.	○	✓	–	○	–	–	Emphasizes sensing–communication–computing synergy, with limited treatment of security/privacy, standards, and long-horizon research challenges.
Magbool et al. [18]	Intelligent metasurface/RIS-assisted ISAC.	–	○	✓	○	–	–	Provides focused metasurface/RIS coverage, but is not intended as a full-field roadmap across theory, networks, prototypes, and standards.
D. Zhang et al. [19]	Recent ISAC evolution and spectrum/platform trends.	○	○	○	○	○	○	Covers recent evolution and selected standardization issues, but does not organize the field around a unified grand-challenge roadmap.
J. Zhang et al. [20]	Intelligent ISAC, learning-assisted design, and system intelligence.	○	✓	○	○	○	–	Connects ISAC with intelligence and learning, but provides limited standards/prototype coverage and no unified challenge/timeline synthesis.
This paper (2026)	Roadmap synthesis across foundations, PHY design, networked ISAC, enabling technologies, applications, prototypes, standardization, grand challenges, and long-horizon directions.	○	✓	✓	✓	○	✓	A broad roadmap rather than a topic-specific survey; several boundary topics are intentionally treated compactly rather than exhaustively.

Note: The table is intended to position the roadmap relative to representative surveys rather than to rank prior work. Coverage labels are based on the depth of treatment within each paper: ✓ indicates dedicated multi-subsection treatment, ○ indicates brief or topic-limited coverage, and – indicates that the topic is outside the paper's main scope. "Std./Proto." denotes standards-facing and/or prototype/measurement-oriented coverage. This paper aims for broad roadmap coverage within its defined RF-ISAC scope, but several areas remain intentionally compact. Low-power ISAC, non-terrestrial network (NTN)-ISAC, detection-theoretic metrics, federated sensing, and digital twins are addressed as boundary or emerging directions rather than as standalone surveys.

distinction among coexistence, cooperation, and co-design, thereby providing a conceptual map for subsequent ISAC research. At roughly the same time, more specialized reviews of DFRC systems cataloged waveform and beamforming strategies for co-located transceivers. These works were foundational, but they were written before networked ISAC, RIS-enabled sensing, and standards-facing design had fully emerged as central themes. By 2023–2024, the survey literature had become more comprehensive. Several reviews began to span information-theoretic foundations, waveform and beamforming design, and selected system-level topics. Even so, these surveys largely predated the recent acceleration of standardization-facing ISAC work and therefore did not address the implications of more recent 3GPP and ETSI developments. Domain-specific reviews also appeared during this period, especially for vehicular ISAC, but they remained narrowly scoped and did not attempt to unify indoor, aerial, infrastructure, and large-scale networked scenarios within a single roadmap.

More recent surveys have expanded the field further, but only piecemeal. A notable recent exception is the broader intelligent-ISAC survey literature, which begins to connect sensing, communication, learning, and system-level intelligence more explicitly, but still stops short of organizing the field around a unified set of grand research challenges and deployment-oriented bottlenecks [20]. RIS-assisted ISAC has received dedicated survey treatment, yet often with stronger coverage of passive RIS than of active RIS or simultaneous transmitting and reflecting RIS (STAR-RIS) configurations [18]. Security-focused reviews have also begun to examine the relationship between threat models, MIMO transceiver vulnerabilities, sensing and communication information leakage, and physical-layer defense mechanisms in perceptive networks [21], [22], [23], [24], [25], [26]. Other recent papers have examined the evolution of ISAC across multiple spectral domains and have incorporated some discussion of standardization trends [19]. However, even these newer surveys generally stop short of formulating a unified research agenda with explicit grand challenges and longer-horizon transformative directions.

Against this backdrop, this paper organizes the existing ISAC literature into a coherent roadmap. It integrates recent 2024–2026 work with the evolving standardization context, identifies a unified set of grand research challenges spanning theory, physical-layer design, networked architectures, security, and cross-layer intelligence, and complements this agenda with longer-horizon transformative directions. It also incorporates prototype and measurement-oriented

studies as a distinct dimension of the literature, connecting theoretical and architectural claims to hardware validation and standards-facing model development [27].

C. CONCEPTUAL FOUNDATIONS OF ISAC

The conceptual foundation of ISAC rests on a simple but important observation: sensing and communication are not merely two functions that happen to share spectrum, but two ways of extracting value from the same transmitted electromagnetic signal. Communication conveys information from a transmitter to an intended receiver, whereas sensing infers properties of the surrounding environment from reflected, scattered, or otherwise transformed versions of that signal. What makes ISAC distinctive, therefore, is not only hardware reuse but also the possibility of pursuing these two objectives within a common signaling and architectural framework.

A useful way to interpret the field is through levels of integration. The first and most conservative level is *radio-communications coexistence* (RCC). In this regime, radar and communication systems remain conceptually distinct and are designed largely independently, even if they occupy adjacent or overlapping spectra. The engineering objective is to manage mutual interference through coordination, separation, null steering, scheduling, or power control. No genuine joint waveform design is attempted, and the two systems continue to operate as separate functionalities that happen to share spectral resources. RCC is therefore best understood as spectrum coexistence rather than true integration.

The second and dominant level is *DFRC*. Here, a single transmitter simultaneously performs radar sensing and communication using a jointly designed waveform, beamformer, or transceiver architecture. The two functions share hardware and spectrum, and often portions of the signal-processing chain, while the tradeoff between them is controlled by explicit design parameters. This is the level at which most present-day ISAC theory is formulated and the level most relevant to current standardization activity. In practical terms, DFRC marks the point at which sensing and communication cease to be merely coordinated and become co-designed. This shift is central to modern ISAC: the same physical signal is intentionally shaped to support both user service and environmental inference.

A speculative third level goes beyond bit-level integration and moves toward *semantic ISAC*. In this regime, the system would operate natively in the domain of task-relevant meaning rather than raw bits and low-level physical observables. Communication would no longer aim simply to transmit data, and sensing would no longer aim simply to estimate parameters such as range, angle, or Doppler with maximum precision. Instead, both functions would be organized around the extraction and delivery of information relevant to a downstream task, such as obstacle avoidance, intent prediction, anomaly detection, or cooperative control. This level remains largely conceptual today, but it is useful because it highlights that ISAC's long-term evolution may bypass the conventional bit-level abstraction altogether.

Semantic ISAC differs from conventional ISAC in the object being optimized. In conventional DFRC or ISAC, the communication side is usually evaluated in terms of rate, reliability, or latency, while the sensing side is evaluated in terms of range, Doppler, angle, CRB, detection probability, localization error, or SMI. In Semantic ISAC, the primary object is instead a task-relevant representation of the scene or network state. The goal may be to communicate whether a road segment is safe, whether a robot should stop, whether an anomaly is present, or which beam and handover decision to take, rather than to reconstruct all raw samples or estimate all physical parameters with equal fidelity. This connects Semantic ISAC to semantic communications, goal-oriented communications, task-oriented networking, and information-bottleneck-style representations. A key open problem is to define semantic fidelity metrics that are operationally meaningful for both sensing and communication, such as decision utility, control loss, task success probability, goal relevance, or action-value preservation. For this reason, Semantic ISAC remains a long-horizon research direction rather than a mature replacement for conventional sensing and communication metrics.

These levels of integration also clarify why ISAC should not be viewed purely through a zero-sum lens. Early treatments often framed the sensing–communication tradeoff as a competition for spectral degrees of freedom: resources devoted to sensing are unavailable to communication, and vice versa. While this captures a real physical constraint, it is incomplete. In many practically important settings, sensing and communication can also benefit one another in ways that fundamentally alter the design picture.

The first such mechanism is *sensing-assisted communication*. Environmental knowledge obtained through sensing—including channel state, user location, scatterer geometry, blockage structure, and mobility information—can be exploited to improve communication performance. For example, sensing-assisted beam management can reduce or even eliminate beam-sweeping latency in mmWave systems by predicting the appropriate beam from the sensed environment rather than discovering it through exhaustive training. More generally, sensing can provide side information that improves resource allocation, mobility prediction, handover support, and robust link adaptation. In this view, sensing is not an added burden on communication resources; it is also a source of communication-side gain.

More concretely, sensing-derived environmental state can be used as side information for network optimization. User locations and velocities can support proactive beam prediction and handover preparation; blockage maps and scatterer geometry can guide multi-connectivity, relay selection, and blockage-aware scheduling; crowd density and traffic flow can inform load balancing and resource partitioning across cells; and environmental dynamics can support predictive link adaptation before channel quality degrades. In networked

ISAC, these mechanisms turn sensing from a passive measurement function into a control input for beam management, scheduling, mobility management, interference coordination, and edge resource allocation. The main challenge is to represent sensing outputs with uncertainty and latency so that network controllers do not overreact to noisy or stale environmental estimates.

The second mechanism is *communication-assisted sensing*. Communication waveforms and communication infrastructure can themselves enhance sensing performance. Reference signals originally designed for channel estimation or synchronization can simultaneously serve as sensing probes, while dense deployments of 5G and 6G base stations naturally create multi-static sensing geometries with improved coverage and localization capabilities. In such settings, the communication network becomes a distributed sensing infrastructure rather than merely a transport fabric [28]. This is particularly important for networked ISAC, where the value of communication lies not only in delivering data to users but also in distributing sensing functionality across space.

Taken together, these observations lead to a more accurate conceptual picture of ISAC. The field is not simply about balancing two competing objectives within a fixed resource budget. It is about identifying when sensing and communication interfere, when they can be jointly optimized, and when each can actively improve the other. RCC, DFRC, and semantic ISAC can therefore be seen as progressively deeper levels of integration, while sensing-assisted communication and communication-assisted sensing reveal why the relationship between the two functions is not purely antagonistic. This perspective provides the conceptual basis for the technical developments discussed in the remainder of the paper.

D. INFORMATION-THEORETIC FOUNDATIONS

The information-theoretic foundation of ISAC is essential because it clarifies what is jointly achievable when communication and sensing are treated not as separate services sharing the same medium, but as coupled inference and transmission tasks over a common physical channel. At this level, the central question is not simply how to allocate power or bandwidth between two functions, but how to characterize the achievable region in the communication rate-sensing fidelity plane, and how that region changes with channel knowledge, signaling structure, and system topology. This perspective has become central to modern ISAC because it provides a rigorous language for expressing the tradeoffs that recur throughout the rest of the field.

A foundational result for Gaussian ISAC channels is the capacity–distortion characterization developed by Xiong et al. [29], [30]. One of the most important insights from this line of work is that the tradeoff has a two-fold structure. The first component is a subspace tradeoff: under a transmit power constraint, allocating spatial or signal-space degrees of freedom to sensing reduces the degrees of freedom available for communication. The second is a deterministic–random tradeoff: deterministic radar-style waveforms are favorable for sensing because they maximize parameter identifiability and Fisher information, whereas random Gaussian codebooks are favorable for communication because they approach channel capacity. Taken together, these two tensions show why ISAC cannot be understood as a trivial superposition of optimal radar and optimal communication designs. In particular, the optimal signaling structure for Gaussian ISAC admits a geometric interpretation in terms of the Stiefel manifold of semi-unitary matrices, a characterization that has been highly influential in subsequent beamforming and DFRC research.

The canonical theoretical formulation of ISAC is closely related to the state-dependent channel model. In this view, the environment or target state acts as a channel state that must be estimated while a communication message is decoded. This connection traces naturally to the Gelfand–Pinsker framework [31]. Building on this perspective, later work characterized capacity–distortion tradeoffs for memoryless ISAC channels under different causality assumptions about state information and extended the analysis beyond single-user links to degraded broadcast configurations [32]. These results are important because they move the ISAC problem from a heuristic tradeoff discussion to formal achievable regions and their converse arguments. Figure 3 provides a conceptual visualization of this idea. The figure is not intended as a universal closed-form boundary for all ISAC systems, but rather as a useful interpretation aid for two persistent theoretical gaps. The first is that the multi-user capacity-distortion region remains largely unknown beyond highly simplified settings. The second is that once finite-blocklength constraints are introduced, the achievable region contracts further, revealing that non-asymptotic penalties may be substantial in latency-sensitive deployments [33], [34], [35]. This figure helps transform an abstract information-theoretic concept into a more concrete picture of how sensing fidelity must be traded against communication rate under realistic constraints.

At the same time, the sensing side of the formulation depends heavily on the chosen metric. Much of the early ISAC literature adopted the Cramér–Rao bound (CRB) as the natural sensing-side performance measure because it offers a clean local lower bound on estimation variance. However, this CRB is a local frequentist metric: it characterizes estimation error around a specific parameter value and depends on regularity conditions such as differentiability of the likelihood function. For many sensing problems, especially Bayesian or distribution-level inference tasks, this is too narrow. Alternative metrics, therefore, play an increasingly important role. *Sensing mutual information (SMI)* measures the information gained about the target-state distribution as a whole, while *estimation rate* formulations characterize the rate at which state uncertainty is reduced by observation [36], [37]. These alternatives do not replace the bound, but they complement it by providing broader views

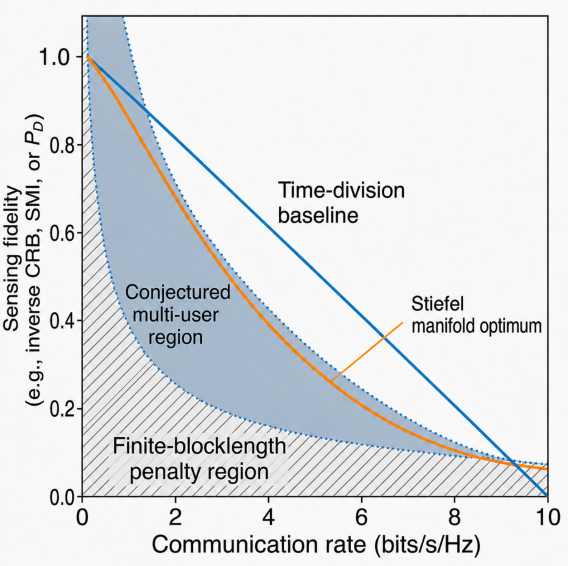


FIGURE 3. Conceptual visualization of the rate–distortion frontier in ISAC. The horizontal axis denotes communication rate, while the vertical axis represents a generic sensing-fidelity surrogate (e.g., inverse CRB, SMI, detection probability, or an equivalent normalized quality metric). The shaded region near the origin indicates a detection-limited regime, where sensing performance is constrained by detectability and ROC behavior rather than estimation variance alone.

of sensing performance and aligning more naturally with probabilistic sensing scenarios.

Detection-theoretic metrics further complement these estimation-oriented measures. The CRB and SMI are meaningful only after a target is detectable; in weak-target, clutter-limited, or low-SNR regimes, the first requirement is to distinguish target presence from absence. This is captured by the probability of detection P_{D}, the probability of false alarm P_{FA}, and the receiver operating characteristic (ROC) curve [38]. In ISAC, the waveform and resource allocation affect both detection and estimation: a design may achieve a favorable estimation bound once the target is detected, yet still be operationally poor if its ROC is weak. Detection probability and false-alarm probability are also natural KPIs for standards-facing infrastructure sensing discussions [39]. Thus, CRB-type estimation metrics, SMI, and ROC-based detection metrics should be treated as complementary pillars of ISAC sensing evaluation.

As ISAC moves from point-to-point links to distributed, multi-user architectures, the relevant theoretical framework shifts from classical single-user information theory to network information theory. The natural next step is to study ISAC over multi-access channels, broadcast channels, interference channels, relay networks, and distributed sensing configurations [40]. Some initial progress has been made in characterizing sensing–throughput tradeoffs for relay-assisted settings and selected multi-user scenarios. However, the general capacity–distortion region for networked ISAC remains largely open. In particular, the simultaneous characterization of sensing and communication limits over these channel models is still missing in its entirety. This gap is not a minor technical detail; it is one of the main reasons why the theoretical foundation of networked ISAC remains significantly less mature than that of single-link DFRC.

Taken together, these results suggest a useful interpretation of the current state of ISAC information theory. The field has already moved beyond the earliest question of whether sensing and communication can coexist at all. It has been established that a principled rate–distortion framework exists, that Gaussian and memoryless models can be analyzed with increasing rigor, and that alternative sensing metrics broaden the scope of what should count as sensing performance. Yet the theory remains incomplete in exactly the regimes that matter most for future systems: short packets, multiple users, and networked infrastructure. For this reason, information-theoretic ISAC should be viewed not as a closed body of results, but as an active and foundational research frontier whose unresolved questions will shape the next phase of the field. A further open issue is fairness in multi-user ISAC. Most current rate–distortion formulations are asymptotic and effectively single-user, whereas practical systems serve users with heterogeneous channel quality, latency constraints, and sensing priorities. Extending the ISAC rate–distortion region to incorporate fairness, differentiated QoS, and shared sensing benefit remains largely open.

III. PHYSICAL-LAYER ISAC DESIGN

The physical layer remains the most mature and analytically developed part of ISAC research because it is where the sensing–communication tradeoff is expressed most directly through waveform structure, beamforming strategy, duplexing mode, and channel interaction. At this layer, the central design question is not simply how to make sensing and communication coexist, but how to make them mutually informative within the same signaling architecture. This objective has motivated a broad body of work on waveform design, spatial processing, full-duplex operation, and ISAC-aware channel representations. Among these topics, waveform design is the most established, both because it admits relatively clean theoretical formulations and because it maps naturally onto existing communication and radar hardware platforms. It is also where some of the most practically important tradeoffs first become visible, including tensions among spectral efficiency, delay–Doppler resolution, robustness to mobility and multipath, hardware complexity, and energy efficiency. This signal-processing centrality is also reflected in recent perspective articles that position waveform design, beamforming, and joint inference as the main analytical backbone of near-term ISAC research [41].

A. WAVEFORM DESIGN FOR JOINT SENSING AND COMMUNICATIONS

Waveform design is the most mature subfield of physical-layer ISAC because the waveform determines, in a first-order

sense, what information is simultaneously made available to the communication receiver and the sensing function. In communication terms, the waveform shapes spectral efficiency, synchronization structure, receiver complexity, and compatibility with existing standards. In sensing terms, it governs range resolution, Doppler resolution, ambiguity structure, and robustness under clutter and mobility. As a result, waveform design is not merely a low-level implementation choice; it is the physical-layer mechanism through which the joint sensing–communication tradeoff is concretized. The literature is centered on four main waveform families: OFDM, orthogonal time-frequency space (OTFS), frequency-modulated continuous wave (FMCW), and hybrid designs that combine communication-oriented and radar-oriented signaling principles [3], [16]. Their comparative strengths are now well understood. OFDM and OTFS align more naturally with communication-centric infrastructures, whereas FMCW retains advantages inherited from classical radar practice, especially in low peak-to-average power ratio (PAPR) operation and mature sensing hardware [42], [43], [44]. Hybrid designs attempt to bridge these two lineages, but often do so at the cost of implementation complexity. Table 2 summarizes the tradeoffs across the most relevant physical-layer metrics, including the energy-efficiency dimension that is important in 6G-oriented evaluations.

OFDM-based ISAC remains the most natural candidate for near-term deployment due to its native compatibility with 5G NR and the potential to reuse existing communication infrastructure, pilot structures, and baseband architectures for sensing [2], [42]. This deployment compatibility is a practical advantage: it lowers the barrier to incremental adoption and makes OFDM especially attractive for standardization-friendly ISAC. However, OFDM also exhibits two well-known weaknesses from a sensing perspective. First, its high peak-to-average power ratio forces the power amplifier to operate with significant backoff, reducing energy efficiency and making hardware-aware implementation more difficult. Second, Doppler resolution is constrained by symbol duration and inter-carrier sensitivity, which limits performance in high-mobility or strongly time-varying environments.

OTFS has emerged as a compelling alternative, especially for high-mobility scenarios, where delay–Doppler-domain signaling offers inherent robustness to time variation [3], [43]. By representing the channel in the delay–Doppler domain rather than the time–frequency domain, OTFS can provide superior Doppler resolution and stronger resilience to doubly selective fading. These features are appealing for ISAC because the sensing task is often naturally expressed in terms of delay and Doppler parameters. Its main limitation is complexity, since delay–Doppler processing, equalization, and practical transceiver implementation remain more demanding than their OFDM counterparts.

FMCW remains dominant in automotive and short-range radar ecosystems because of its low PAPR, strong ranging capability, and highly mature hardware support [44]. From a sensing perspective, it is particularly attractive because it provides high range resolution with favorable energy efficiency and well-understood analog front-end behavior. From a communication perspective, however, it is much less natural. Its spectral efficiency is limited relative to multicarrier waveforms, and its signaling structure is not well aligned with native integration into cellular communication standards.

Hybrid designs aim to combine the advantages of communication-driven waveforms such as OFDM or OTFS with radar-favorable properties of FMCW-like sensing [3], [16]. Their promise is clear, but so is the implementation challenge: hybrid designs often increase transceiver complexity, complicate receiver synchronization and resource allocation, and may require hardware architectures that are neither as mature as FMCW nor as deployment-compatible as OFDM.

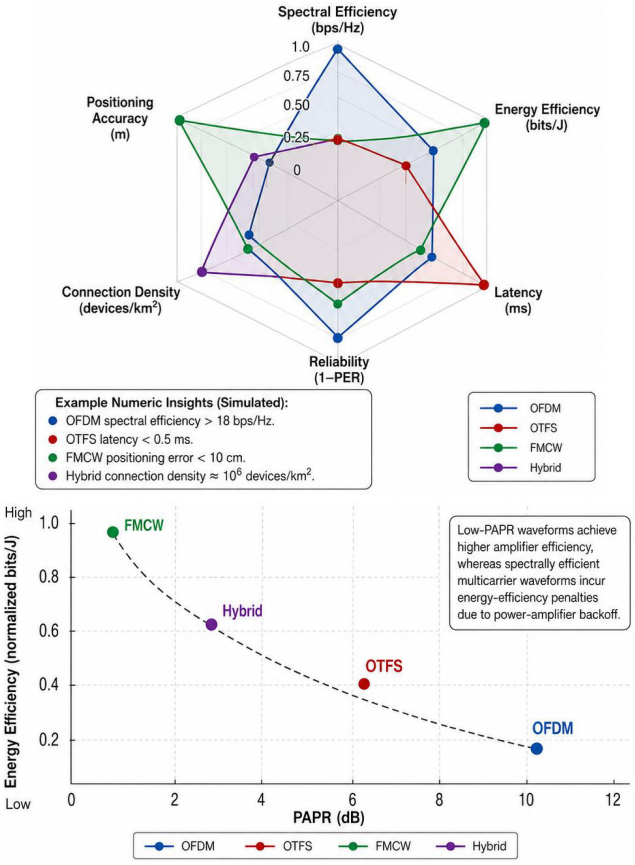


FIGURE 4. Conceptual comparison of four representative ISAC waveform candidates—OFDM, OTFS, FMCW, and hybrid designs—across key communication and sensing metrics. The comparison highlights qualitative trends rather than deployment-independent rankings; the relative ordering depends on bandwidth, frame structure, mobility, synchronization, hardware constraints, and receiver processing.

An interpretation of the design space is that no single waveform dominates across all relevant metrics. OFDM is strongest in near-term deployment compatibility and communication integration; OTFS is strongest in delay–Doppler robustness and high-mobility operation; FMCW is strongest in sensing efficiency and hardware maturity; and hybrid

TABLE 2. Qualitative comparison of representative ISAC waveform candidates across key communication and sensing performance metrics. Entries summarize common trends reported in the recent ISAC literature.

Metric	OFDM	OTFS	FMCW	Hybrid	Indicative Ref.
Range Resolution	Moderate	Moderate–High	High	Moderate	[42]–[45]
Doppler Resolution	Low	High	Moderate	Moderate	[42]–[45]
PAPR	High	High	Low	Low–Medium	[42]–[44]
Spectral Efficiency	High	High	Low	Moderate	[3], [16], [42], [43], [45]
Multipath Robustness	Moderate–High	Very High	Low–Moderate	Moderate	[42], [43], [45]
Hardware Complexity	Low	High	Low–Moderate	Moderate–High	[3], [16], [42]–[44]
5G/6G Compatibility	Native	Planned	Partial	Partial	[2], [3], [16], [39], [46]
Energy Efficiency	Moderate	Moderate	High	Moderate–High	[16], [42], [44], [45]
ISAC Suitability (near-term)	High	Moderate–High	Moderate	Moderate	[2], [3], [16], [42], [45]

Note: Ratings are qualitative and scenario-dependent. They summarize common trends under representative RF-ISAC assumptions and are not strict quantitative rankings. Range and Doppler resolution depend on bandwidth, coherent processing interval, carrier frequency, frame structure, mobility regime, synchronization, receiver processing, and hardware constraints. Energy efficiency depends strongly on PAPR, power-amplifier operating point, converter resolution, and duty cycle. PAPR = peak-to-average power ratio.

designs occupy the space between these extremes. Figure 4 provides a compact qualitative comparison of these waveform families across the main communication and sensing dimensions, while Figure 5 highlights the important tradeoff between PAPR and energy efficiency. The latter is significant because it illustrates a central tension for 6G ISAC: waveforms that are attractive from the perspective of communication spectral efficiency may be less attractive from the perspective of amplifier efficiency and deployment-level energy cost.

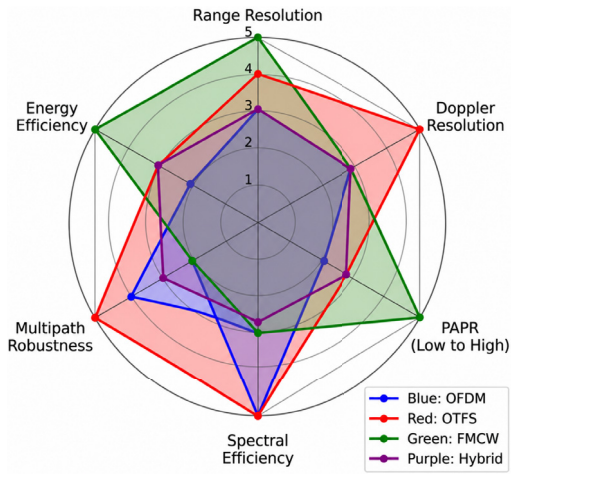


FIGURE 5. Illustrative tradeoff between PAPR and energy efficiency for representative ISAC waveform families. The figure is conceptual and highlights a common hardware-efficiency trend: low-PAPR radar-oriented designs such as FMCW may remain attractive from an amplifier-efficiency perspective, even when communication-oriented waveforms offer better spectral efficiency.

The broader implication is that waveform design for ISAC should not be framed as a search for a universally optimal signaling format. This point is reinforced by recent monostatic OFDM-ISAC studies, which show that the relevant trade space is inherently multi-dimensional, spanning modulation-dependent ambiguity behavior, sensing sidelobe control, and the choice between concurrent and time-sharing transmission strategies rather than a single scalar optimum [45]. Rather, it should be understood as a structured design problem in which different application regimes, hardware constraints, and standardization requirements favor different points on a multi-dimensional tradeoff surface. Near-term cellular ISAC is likely to remain dominated by OFDM-based approaches due to compatibility and reuse. High mobility and delay–Doppler-centric scenarios may increasingly favor OTFS. Automotive and dedicated sensing deployments will continue to rely heavily on FMCW-derived designs. Hybrid approaches may become more important as hardware matures and as the need grows for waveforms that explicitly balance communication efficiency, sensing quality, and energy-aware implementation. As a result, waveform design remains central not only because it is the most mature physical-layer topic in ISAC, but because it continues to define the practical boundary between what is theoretically attractive and what is actually deployable. For OFDM-based ISAC, the cyclic prefix and guard-interval structure also influence the effective ambiguity behavior by introducing frame-dependent range sidelobes and Doppler artifacts, especially when sensing is performed using communication-oriented reference structures rather than dedicated radar probing sequences [45].

B. BEAMFORMING AND SPATIAL SIGNAL PROCESSING

Beamforming and spatial signal processing are central to physical-layer ISAC because spatial degrees of freedom determine how effectively a system can simultaneously serve communication users and sense the environment. Both functions fundamentally rely on controlling the transmitted spatial signature and extracting structure from the received angular response. In conventional designs, communication and sensing are often treated as separate spatial tasks, with

one beam dedicated to data transmission and another reserved for environmental probing. Recent ISAC results, however, suggest a more unified interpretation: the spatial pattern itself can be jointly designed so that the same transmitted data streams are informative for both communication and sensing. This shift from resource separation to unified spatial design is one of the defining insights of DFRC beamforming.

A key finding of the downlink ISAC beamforming literature is that the optimal transmit beamformer is not, in general, a simple superposition of a communication beamformer and a dedicated sensing beam [47], [48]. Instead, the sensing function can be embedded directly into the communication-bearing signal when the beamformer is jointly optimized. In other words, the data streams themselves carry sensing information through their spatial structure, provided that the beamformer is designed to shape the transmitted field in a way that is simultaneously useful for user service and target illumination. This insight is important because it shows that communication and sensing need not always compete for orthogonal resources; under appropriate design, they can be unified at the beamforming layer. The resulting optimization is inherently multi-objective, since the beamformer must balance communication metrics, such as rate or SINR, with sensing metrics, such as beampattern fidelity, localization sensitivity, or target illumination quality.

This unified perspective also clarifies why beamforming is more than an extension of classical multi-user precoding [48]. In communication-only systems, the transmit beamformer is typically optimized to separate users and maximize rate subject to power and quality-of-service constraints. In ISAC, the same spatial degrees of freedom must also encode information about the environment, which introduces additional design objectives that may not align with conventional communication optima. The beamformer must therefore mediate a Pareto tradeoff between communication efficiency and sensing utility. This tradeoff is often visualized through a rate–sensing frontier, where improving one function beyond a certain point inevitably degrades the other. The practical goal is not to eliminate this tension, but to identify operating points where both functions benefit from unified spatial design rather than from strict partitioning.

Full-duplex operation is attractive in this context because it allows simultaneous transmission and reception on the same frequency band, thereby enabling monostatic sensing while maintaining communication service. From an ISAC perspective, this is desirable: the transmitter can illuminate the environment and receive echoes without requiring separate time or frequency resources, thereby improving spectral efficiency and simplifying the integration of sensing into a communication platform. However, the full-duplex operation introduces self-interference from the transmit chain into the receive chain, often tens of decibels stronger than the desired echo or uplink signal. This makes self-interference cancellation (SIC) a critical enabler of full-duplex ISAC. From a hardware perspective, this challenge is severe because ADC dynamic range, front-end linearity, and residual nonlinear distortion can easily dominate the useful echo or uplink component unless SIC is co-designed with the transceiver chain [49]. SIC in ISAC cannot be treated as a communication-only subsystem, because the transmitted waveform and beamformer jointly determine the structure of the self-interference. The cancellation architecture must be co-designed with the ISAC waveform and spatial processing strategy, rather than appended as a front-end block.

Full-duplex ISAC also differs from communication-only full duplex because the receiver must cancel the direct leakage from the transmitter without suppressing weak target echoes that are structurally related to the transmitted waveform. Residual self-interference may include linear leakage, power-amplifier nonlinearities, phase-noise-induced spreading, IQ imbalance, antenna coupling, and quantization effects due to the limited ADC dynamic range. Analog cancellation is needed to prevent receiver saturation, while digital cancellation must preserve the delay–Doppler structure of the desired echo. Beam-domain suppression can help, but it interacts with the sensing beampattern and may create blind zones. As a result, full-duplex ISAC requires joint design of the waveform, transmit beamformer, analog cancellation, digital SIC, and sensing receiver, rather than treating SIC as a separate communication front-end module.

Figure 6 illustrates three representative beamforming regimes that capture the evolution of the field. The first is the legacy separate-beam approach, in which communication and sensing use different beams and are often assigned separate spatial resources. This architecture is conceptually simple and compatible with earlier coexistence-oriented designs, but it is suboptimal because it sacrifices spatial degrees of freedom and limits the possibility of mutual benefit between the two functions. The second is DFRC joint beamforming, which represents the current state of the art in many single-cell ISAC formulations. Here, a single jointly designed beam pattern supports both communication and sensing, and the design objective is to select an operating point on the communication–sensing Pareto frontier. The third is multi-cell cooperative beamforming, which extends the DFRC idea to distributed and networked settings. In this regime, multiple access points coordinate their spatial transmissions to jointly serve users and sense targets, enabling coverage extension, improved localization, and interference-aware sensing through spatial diversity and cooperation.

Beamforming and spatial processing provide one of the clearest examples of how ISAC departs from the conventional view that sensing and communication must be overlaid as parallel functionalities. At the spatial level, the two functions can often be fused more deeply than is possible in time- or frequency-based partitioning. This is why beamforming remains such a central research topic: it offers a concrete mechanism through which unified signaling can translate into performance gain. At the same time, it exposes the main

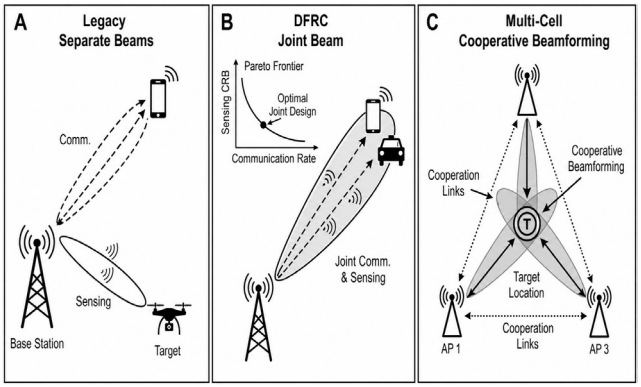


FIGURE 6. Evolution of ISAC beamforming architectures. The legacy approach uses separate communication and sensing beams, which is simple but inefficient. DFRC joint beamforming unifies the two functions through a spatial design. Multi-cell cooperative beamforming extends this concept to distributed networked ISAC, where multiple access points coordinate to improve coverage, target localization, and sensing resolution.

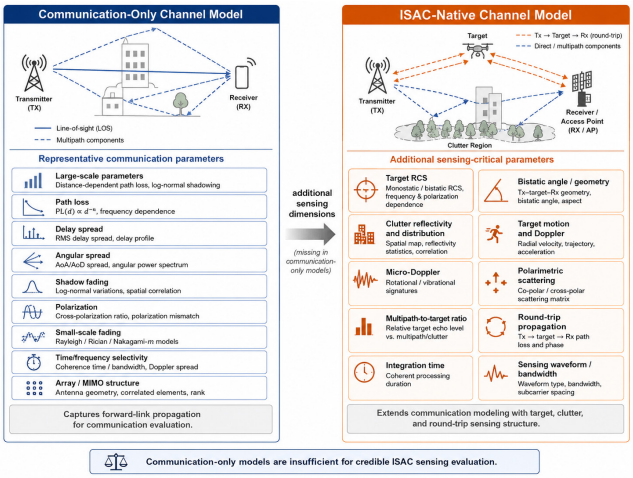


FIGURE 7. Conceptual comparison between communication-only channel modeling and an ISAC-native sensing channel model. Communication models primarily characterize forward-link multipath statistics, whereas ISAC-native models must also capture target scattering, Doppler and micro-Doppler, clutter structure, and round-trip propagation geometry within a mutually consistent environmental representation.

challenges that remain open, including the characterization of communication–sensing Pareto structure, SIC for full-duplex ISAC, and the extension from single-cell joint beamforming to large-scale cooperative multi-cell architectures.

C. ISAC-NATIVE CHANNEL MODELING

Modern wireless channel models, including geometry-based stochastic models such as 3GPP TR 38.901 [50], already capture rich forward-link communication phenomena, including delay spread, angular spread, shadow fading, polarization, mobility-induced Doppler, blockage, and cluster evolution. The limitation is not that these models lack environmental structure. Rather, their primary purpose is to model the transmitter-to-receiver communication channel. ISAC requires additional sensing-native layers that answer different questions: how a target scatters energy, how its RCS varies with aspect angle and frequency, how clutter statistics shape detection thresholds, how round-trip or bistatic propagation changes echo power, and how micro-Doppler, target fluctuation, and ghost targets affect inference. These quantities are not generally represented as first-class objects in communication-only models, yet they determine detection probability, false-alarm behavior, localization accuracy, and multi-static fusion consistency. Recent standardization-oriented experimental work has begun to address this gap, but a broadly adopted ISAC-native channel model is still missing [27]. This gap is especially visible when comparing sensing requirements with the much richer measurement and modeling ecosystem already developed for mmWave communication channels, which captures forward-link multipath well but does not natively represent target scattering, clutter structure, or round-trip sensing geometry [51], [52].

Figure 7 illustrates this distinction by contrasting forward-link communication modeling with an ISAC-native model that explicitly represents target scattering, Doppler, clutter, and round-trip geometry. To make this distinction concrete, a conventional communication channel can be expressed as

$$h(\tau, \theta_{\mathrm{AoD}}, \theta_{\mathrm{AoA}}) = \sum_{p=1}^{P} \alpha_p \, \delta(\tau - \tau_p) \, \delta(\theta_{\mathrm{AoD}} - \theta_{\mathrm{AoD},p}) \times \delta(\theta_{\mathrm{AoA}} - \theta_{\mathrm{AoA},p}). \quad (1)$$

where α_p is the complex path gain, τ_p is the path delay, and $\theta_{\mathrm{AoD},p}$ and $\theta_{\mathrm{AoA},p}$ denote the departure and arrival angles of path p. Such a model is appropriate when the goal is to characterize the stochastic forward propagation channel for communication. By contrast, sensing requires more information. Once a propagation path interacts with a target, the model must represent the target's radar cross section (RCS), its angular and frequency dependence, its motion-induced Doppler signature, and the clutter background statistics. These quantities are not optional refinements; they determine detection probability, estimation accuracy, false-alarm behavior, and the structure of multi-static sensing gain.

Several missing ingredients are important. First, the bistatic or monostatic RCS must be represented rather than absorbed into an undifferentiated complex path coefficient. For communication, the path gain α_p is often sufficient; for sensing, the RCS must be modeled separately because it determines echo strength and varies sharply with aspect angle. Second, Doppler must be treated as a signal rather than a nuisance. Communication models typically retain Doppler only when it degrades channel coherence or induces inter-carrier interference, whereas sensing requires Doppler and micro-Doppler as primary observables for velocity estimation and target classification. Third, clutter must be treated as a structured random field rather than merely as interference. Ground, vegetation, buildings, rain, and

indoor scatterers form a background whose statistics directly influence threshold setting and detection reliability. Finally, the propagation law itself differs: monostatic sensing obeys round-trip scaling proportional to $1/r^4$, and bistatic sensing scales as $1/(r_1^2 r_2^2)$, whereas communication models typically adopt a forward-link power law of the form $1/r^\gamma$. Using the wrong propagation structure can bias not only link budgets but also sensing range prediction and localization behavior.

The quantitative gap is clear. A communication-oriented, geometry-based stochastic model may specify delay spread, angular spread, shadow fading, polarization, and cluster evolution, yet still omit many sensing-relevant parameters that dominate target detectability and estimation fidelity. Table 3 summarizes a set of sensing-critical quantities that are absent from standard communication-only models but required for credible ISAC evaluation. These parameters are not merely implementation details. They directly affect the achievable sensing signal-to-noise ratio, clutter-limited detection threshold, target separability, ghost-target risk, and the consistency of distributed multi-static fusion. This motivates a different formulation of what a channel model must answer. An ISAC-native model should support two mutually consistent query types. The first is the conventional communication query: given a transmitter at location $\mathbf{x}_t$ and a receiver at location $\mathbf{x}_r$, what is the distribution of the received signal $y(t)$ for a transmitted signal $x(t)$? The second is the sensing query: given a transmitter at $\mathbf{x}_t$, a receiver at $\mathbf{x}_r$ (which may be co-located or distributed), and a target at location $\mathbf{x}_\tau$ with scattering characteristics σ, what is the expected target-echo SNR and what is the distribution of the clutter-plus-noise background? These queries must be answered using the same underlying physical environment. That consistency is largely absent in current practice, where communication and radar models are often built on incompatible environmental abstractions.

A practical way forward is to extend geometry-based stochastic communication models with explicit sensing layers. In particular, an ISAC-native extension should include: a *target layer*, containing discrete targets with position, velocity, frequency- and angle-dependent RCS, fluctuation model, and micro-Doppler signature; a *clutter map*, partitioning the environment into regions with specified reflectivity and clutter distributions; *round-trip propagation rules*, computing delay, Doppler, and received power for transmitter–target–receiver triplets; and *bistatic extensions*, allowing transmitter–receiver pairings for cell-free, cooperative, and networked sensing. Under such a model, the same environment generates both the multipath communication channel and the structured sensing scene, making communication and sensing evaluation consistent rather than artificially stitched together. More concretely, for a transmitter–target–receiver triplet, the sensing channel should compute the round-trip delay, Doppler, and received power according to

$$\tau = \frac{d_{\text{tx}} + d_{\text{rx}}}{c}, \tag{2}$$

$$f_D = \frac{\left(\mathbf{v}_\tau \cdot \hat{\mathbf{r}}_{\text{tx}} + \mathbf{v}_\tau \cdot \hat{\mathbf{r}}_{\text{rx}}\right) f_c}{c}, \tag{3}$$

and

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 d_{\text{tx}}^2 d_{\text{rx}}^2 L}, \tag{4}$$

where d_{tx} and d_{rx} denote the transmitter-to-target and target-to-receiver distances, respectively, and L captures propagation or system losses. These expressions make explicit what communication-only models suppress: sensing depends on geometry, target scattering, and round-trip propagation in ways that cannot be reduced to forward-link fading.

The challenge is not only conceptual, but empirical. Channel models cannot mature on theory alone; they require measurement campaigns. At present, no publicly available dataset exists that simultaneously characterizes communication multipath, target RCS, and clutter behavior in the same physical environment and frequency band. Existing standardization-oriented efforts provide an important starting point, but they do not yet constitute a comprehensive multi-scenario ISAC-native modeling baseline [27], [50]. What is needed are coordinated measurement campaigns spanning both sub-6 GHz and mmWave bands across representative scenarios such as indoor factories, urban street canyons, highway vehicular settings, offices, and aerial-drone surveillance. Such campaigns should include calibrated measurements of target aspect dependence, clutter statistics, communication multipath, and bistatic geometry under the same environmental conditions. A public multi-scenario, multi-band ISAC dataset would provide a reproducible baseline for model development and standards-facing evaluation. This is especially important because model realism will directly determine whether future ISAC evaluations remain comparable across papers, simulators, and vendors.

The broader implication is that ISAC-native channel modeling must be treated as enabling infrastructure for the field rather than as a downstream documentation task. Without realistic models, information-theoretic studies remain trapped in abstract Gaussian assumptions, distributed sensing analyses cannot capture bistatic echo structure, and network-level resource-allocation studies cannot represent interference coupling through clutter and target echoes. Future progress in ISAC depends not only on better algorithms, but also on realistic environmental abstractions, agreed parameter sets, and measurement-validated baselines. Without that foundation, results across the field remain difficult to compare, and deployment-oriented conclusions remain vulnerable to hidden optimism in the underlying channel assumptions.

D. WAVEFORM, ARRAY, AND METRIC TRADEOFFS IN PHYSICAL-LAYER ISAC

The preceding subsections make clear that physical-layer ISAC design is inherently multi-objective. No waveform, beamforming strategy, or channel abstraction is uniformly optimal across communication rate, sensing fidelity, energy

TABLE 3. Representative sensing-relevant parameters not generally modeled as first-class quantities in communication-only channel models. The listed ranges are illustrative and scenario-dependent rather than universal bounds, and highlight the additional modeling requirements of ISAC-native evaluation.

Missing Parameter	Symbol	Typical Range	Impact on Sensing Performance
Target RCS (monostatic)	σ_{mono}	-20 to $+30$ dBsm	Detection probability, echo SNR
Target RCS (bistatic)	$\sigma_{\mathrm{bi}}(\theta)$	Angle-dependent	Multi-static fusion gain, target visibility
RCS angular correlation distance	$\Delta\theta_{\mathrm{corr}}$	0.5°–10°	Track continuity and aspect sensitivity
Clutter reflectivity	σ^0	-50 to -10 dB	False-alarm behavior, background power
Clutter distribution shape	ν	0.1–10	CFAR threshold design and detection robustness
Clutter decorrelation time	T_c^{clutter}	10 ms–1 s	Moving-target indication and temporal filtering
Micro-Doppler modulation	$f_{\mu D}$	10 Hz–10 kHz	Target classification and motion signatures
Target fluctuation model	Swerling I–IV	N/A	Detection probability variation
Polarimetric scattering matrix	$\mathbf{S}$	Complex-valued	Target identification and material dependence
Multipath-to-target ratio	MTR	-20 to $+10$ dB	Ghost-target risk and echo ambiguity
Bistatic angle	β	0°–180°	RCS variation and geometry dependence
Doppler ambiguity interval	$f_{D,\max}$	application-dependent	Velocity unwrapping and Doppler observability
Range resolution cell size	$\Delta R = c/(2B)$	cm to m	Minimum separable target distance
Angular resolution	$\Delta\theta = \lambda/(D\cos\theta)$	Degrees	Minimum separable angle
Radar horizon / coverage constraint	R_{horizon}	scenario-dependent	Coverage prediction and sensing reach
Coherent synchronization error	Δt	1 ps–10 ns	Phase alignment and coherent multi-static gain

efficiency, implementation complexity, and deployment compatibility. Instead, physical-layer ISAC should be understood as a structured trade space in which waveform choice, array architecture, and performance metrics are tightly coupled. The central design problem is therefore not to identify a universally superior signaling strategy, but to determine which combination of waveform, spatial processing, and modeling assumptions is most appropriate for a given operating regime.

Concrete Pareto mechanisms arise at several levels. In time-domain partitioning, allocating more symbols to sensing improves coherent integration and Doppler resolution but reduces data throughput and may increase communication latency. In frequency-domain partitioning, dedicating more bandwidth to sensing improves range resolution but reduces communication spectral resources. In waveform design, deterministic or low-PAPR radar-like signals may improve ambiguity behavior and amplifier efficiency, while random or higher-order communication waveforms may more closely approach communication capacity. In spatial processing, narrow beams improve user SINR and target illumination when the geometry is known, but broader beams may be needed for search, tracking under uncertainty, or scene awareness. In full-duplex ISAC, simultaneous operation improves spectral reuse, but the required self-interference cancellation and receiver dynamic range can degrade weak-echo detection. These examples show why the sensing–communication frontier depends on the operating regime, sensing metric, channel model, and hardware constraints.

At the waveform level, OFDM remains the most natural option for infrastructure-compatible ISAC because of its alignment with standards and implementation maturity, while OTFS is attractive in high-mobility and delay-Doppler-centric regimes, and FMCW remains compelling where sensing efficiency and radar-hardware maturity dominate. Hybrid approaches occupy the space between these extremes, but usually at the cost of higher architectural complexity. At the array level, narrow beams improve communication SINR and target illumination in favorable geometries, but may reduce sensing robustness when target location is uncertain or when broader scene observability is needed. Distributed and cooperative arrays can improve localization accuracy and geometric diversity, but only when synchronization, fronthaul, and interference constraints are sufficiently well controlled. These design choices are inseparable from the choice of performance metric. Communication performance is usually expressed in terms of rate, error probability, or spectral efficiency, whereas sensing performance may be quantified by CRBs, detection probability, estimation error, ambiguity behavior, or SMI. The same design may therefore be ranked quite differently depending on the objective function. This implies that physical-layer ISAC

cannot be reduced to a single scalar optimization target without obscuring the underlying tradeoff structure.

A final and often underappreciated issue is the tension between analytical convenience and physical realism. Simplified Gaussian models, idealized hardware assumptions, and communication-only channel abstractions often enable tractable analysis, but they may substantially overstate sensing performance once target scattering, clutter, round-trip propagation, phase noise, amplifier nonlinearity, and self-interference are accounted for. For this reason, the most practically relevant physical-layer ISAC designs are unlikely to be those that optimize a single dimension most aggressively. Rather, they will be those that achieve balanced performance across communication efficiency, sensing utility, hardware implementability, and deployment compatibility.

IV. NETWORKED AND DISTRIBUTED ISAC

ISAC becomes more powerful—and more difficult to realize—when it moves beyond a single link or co-located transceiver to a network of distributed access points, cooperative receivers, and shared sensing resources. This viewpoint aligns with the emerging notion of sensing as a service in perceptive mobile networks, where distributed infrastructure is designed not only to carry traffic but also to expose sensing capabilities as a network-level service abstraction [28]. In this regime, ISAC is no longer only a physical-layer co-design problem; it is also a system-level problem shaped by architecture, fronthaul constraints, synchronization limits, inter-cell interference, and large-scale resource coupling. This section examines how sensing and communication performance evolve when ISAC is deployed as a coordinated networked infrastructure, with emphasis on the architectural choices, implementation bottlenecks, and scaling laws that determine whether distributed cooperation yields meaningful gains in coverage, localization accuracy, and spectral efficiency. This networked viewpoint also sets the stage for privacy-preserving distributed learning architectures, including federated sensing in Section VII-D, and for the NTN-ISAC extension discussed in Section VI-D.

A. CELL-FREE AND MULTI-STATIC ISAC ARCHITECTURES

Conventional cellular networks are organized around fixed cell boundaries, with each base station primarily serving users located within its own coverage area. While this architecture is effective for communication, it is inherently limited for sensing, since each base station typically performs sensing in isolation using only its own transmissions and receptions, and sensing measurements are not systematically fused across neighboring cells. As a result, sensing performance is constrained by the geometry, visibility, and signal-to-noise ratio (SNR) of a single site, or at most a small number of loosely coordinated sites.

Cell-free ISAC departs fundamentally from this model by removing the notion of a privileged serving cell. Instead, a large number of distributed access points (APs), connected via fronthauls to one or more central processing units, jointly serve users and sense targets within a common coverage region [53], [54]. From a sensing perspective, this transforms the network into a distributed multi-static radar, in which many transmit–receive pairs cooperate to observe the same scene. This distinction is important: cell-free ISAC is not simply a dense small-cell deployment with limited coordination. In a small-cell system, each AP still performs largely independent scheduling and local inference. In a cell-free architecture, by contrast, the distributed APs act as a unified sensing-and-communication infrastructure, and a user or target may be jointly illuminated, observed, and processed by all APs within effective range. Figure 8 illustrates the shift in architecture from local cellular sensing to cooperative cell-free multi-static sensing.

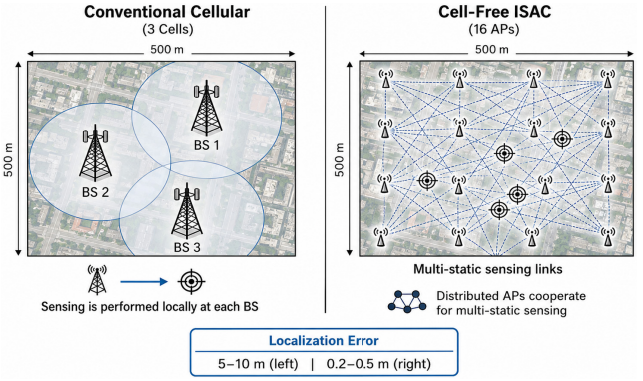


FIGURE 8. Comparison between conventional cellular sensing and cell-free ISAC over the same coverage area. In the conventional architecture, sensing is performed locally at individual base stations, limiting geometric diversity and localization accuracy. In the cell-free architecture, distributed access points cooperate via multi-static sensing links and shared processing, thereby improving localization performance.

The main advantage of this architecture is sensing diversity. In a conventional cellular system, a target at location $\mathbf{p}_\tau$ is typically sensed only by the nearest base station, or perhaps by two BSs during coordination or handover. In a cell-free deployment with K distributed APs, the same target can be observed through many transmitter–receiver combinations, yielding a much richer sensing geometry. Under coherent processing, the effective sensing SNR can be expressed as

$$\mathrm{SNR}_{\mathrm{CF}} = \frac{P_{\mathrm{tx}}\lambda^2}{(4\pi)^3\,(N_0 + I_{\mathrm{total}})} \sum_{i=1}^{K}\sum_{j=1}^{K} \frac{G_i G_j \sigma_{ij}(\theta_i, \theta_j)}{d_{i\tau}^2 d_{j\tau}^2}, \quad (5)$$

where σ_{ij} denotes the bistatic RCS associated with transmitter AP i and receiver AP j, and $d_{i\tau} = \|\mathbf{p}_i - \mathbf{p}_\tau\|$ is the distance between AP i and the target. Equation (5) highlights the central benefit of cell-free sensing: instead of relying on a single monostatic echo path, the network can combine information across many distributed propagation paths.

The resulting gain arises through several complementary forms of diversity. First, *transmit diversity* enables multiple APs to illuminate the same target, thereby increasing the aggregate reflected energy compared to a single-base station

architecture. Second, *receive diversity* enables multiple APs to observe the reflected signal, improving robustness and increasing detection reliability. Third, *geometric diversity* arises because the target is observed from many bistatic angles, thereby reducing sensitivity to aspect-dependent RCS fluctuations. Finally, when different APs operate across different frequency resources or bands, *frequency diversity* can further improve robustness by averaging over frequency-dependent target and channel effects.

A simple back-of-the-envelope example illustrates the scale of the possible gain under idealized assumptions. Consider $K = 16$ APs uniformly distributed over a 500 m $\times$ 500 m area, with a target near the center, carrier frequency $f_c = 3.5$ GHz, transmit power $P_{\text{tx}} = 23$ dBm per AP, antenna gain $G = 5$ dBi, and a nonfluctuating target RCS of $\sigma = 1\,\text{m}^2$. Assume a fixed coherent processing interval, negligible clutter, no dominant cross-sensing interference, calibrated receiver noise figure, and sufficiently accurate synchronization for coherent combining over the selected AP set. Under these assumptions, coherent combining over K comparable observations can provide an array-like gain on the order of $10\log_{10} K \approx 12$ dB, while additional bistatic geometric diversity may improve detectability when the target is observed from favorable angles. The exact SNR depends strongly on RCS fluctuation, AP-target distances, bandwidth, receiver noise figure, interference, synchronization error, and fusion architecture. Thus, the example should be interpreted as an illustrative order-of-magnitude calculation rather than a deployment-independent prediction.

Cell-free ISAC should be viewed not merely as a coverage-enhancement strategy, but as a new sensing architecture. By replacing isolated monostatic sensing with coordinated multi-static observation, it creates new opportunities for robust detection, improved localization accuracy, and wider-area situational awareness. At the same time, these gains come at the cost of greater fronthaul demand, tighter synchronization requirements, and increased processing complexity, which motivate the discussion in the following subsections. A related but distinct architecture is passive ISAC, in which a receiver exploits signals of opportunity from communication transmitters that it does not directly control. Unlike cell-free or fully cooperative multi-static ISAC, passive ISAC cannot jointly optimize the sensing waveform at the transmitter and must instead operate under waveform-agnostic constraints, uncertain synchronization, and limited knowledge of transmitter scheduling. This makes passive ISAC less flexible than co-designed DFRC or cell-free architectures, but relevant in heterogeneous deployments where communication infrastructure is available but not fully cooperative.

B. FRONTHAUL, CENTRALIZATION, AND DISTRIBUTED PROCESSING

The sensing gains of cell-free and multi-static ISAC are realized only if distributed observations can be transported, fused, and processed across the network. This makes fronthaul and processing architecture central design variables rather than secondary implementation details. In particular, the value of distributed sensing depends on how much information each AP forwards, where fusion is performed, and whether the resulting latency and communication overhead remain compatible with the sensing task. Accordingly, networked ISAC is governed by a three-way tradeoff among information fidelity, transport cost, and processing complexity.

The first bottleneck is fronthaul. In a cell-free architecture, each AP must forward some representation of its local observation to a central processor or to neighboring APs. The most information-rich option is to transmit raw in-phase and quadrature (IQ) samples, which preserves full phase information and supports coherent multi-static fusion. For a bandwidth of 100 MHz, a 12-bit analog-to-digital converter, and a 1 ms coherent processing interval, the required fronthaul rate is on the order of

$$R_{\text{IQ}} \approx 2Bb_{\text{ADC}}, \tag{6}$$

which yields approximately 2.4 Gbps per AP. Such rates are manageable only in small-scale prototypes or provisioned deployments. Compression can reduce this burden, but the design of the compression operator becomes part of the sensing problem itself, since overly aggressive quantization or transform-domain truncation can destroy the weak echoes and phase relationships needed for coherent processing.

At the opposite extreme, an AP may forward only local detection reports or local parameter estimates such as range, Doppler, and angle. This reduces fronthaul dramatically, often to the Mbps regime, and scales far better with AP density. However, it also sacrifices the very information that makes cell-free sensing attractive: coherent combining across APs, weak-target recovery, joint data association, and consistent uncertainty propagation. A particularly important and somewhat counterintuitive point is that forwarding a fully formed range–Doppler map is not necessarily fronthaul-efficient. For a representative 512 $\times$ 512 map with 16-bit entries and a 1 ms update interval, the required data rate can exceed that of raw IQ forwarding, because the map is an overcomplete intermediate representation rather than a compressed sufficient statistic. As a result, fronthaul design is not simply a matter of reducing bitrate; it is a matter of selecting the representation that best preserves decision-relevant information within a given transport budget.

These transport constraints give rise to distinct processing architectures. In a *centralized* architecture, all AP observations are delivered to a CPU, which performs joint sensing and fusion. This approach offers the best sensing performance because it preserves full network-wide coherence and enables global inference over all transmitter–receiver pairs. Its weaknesses are clear: high fronthaul demand, a heavy computational bottleneck at the CPU, and aggregation latency that can become problematic as the number of APs grows. Centralized processing is attractive when the number of APs is moderate, and the application demands the highest possible localization accuracy or weak-target detectability.

A *hierarchical* architecture reduces this burden by introducing intermediate fusion points. APs are grouped into clusters, with local cluster processors performing partial fusion before forwarding refined outputs to a higher-level controller. This structure lowers fronthaul load relative to a fully centralized design while preserving much of the gain from local coherent fusion. Hierarchical processing is particularly appealing when the sensing scene exhibits geographic locality or when the network naturally partitions into regions with strong intra-cluster coupling and weaker inter-cluster interactions. In such cases, most of the coherent gain can be captured locally, while inter-cluster fusion can be performed at a lower rate using tracks, likelihood summaries, or compressed sufficient statistics.

In a *distributed* architecture, APs do not send raw observations to a central CPU. Instead, they exchange messages and iteratively update local beliefs or estimates through distributed inference. Representative mechanisms include distributed gradient descent on a joint likelihood, belief propagation on a factor graph, and alternating direction method of multipliers (ADMM) for consensus-based optimization. These methods are attractive because they shift the burden away from fronthaul-intensive raw-data transport and eliminate a single fusion bottleneck. However, they introduce a different constraint: convergence. The number of message exchanges required to approach the performance of centralized fusion depends strongly on signal-to-noise ratio, target density, graph connectivity, and the non-convexity of the sensing problem. In particular, measurement association and multipath ambiguity can make the effective likelihood highly non-convex, so that simple convergence guarantees from convex optimization no longer apply. Distributed processing improves scalability but only at the cost of increased algorithmic complexity and potentially significant sensing latency.

A *hybrid* architecture combines these approaches adaptively. Under favorable conditions, such as strong targets or fronthaul congestion, APs may forward local detections and rely on non-coherent track fusion. Under more demanding conditions, such as weak targets or fine localization tasks, selected APs may switch to raw or lightly compressed IQ forwarding to allow the CPU to perform coherent joint processing. This adaptive viewpoint is especially important for practical ISAC because it recognizes that the optimal architecture is operating-point-dependent rather than universal.

The resulting design tradeoff can be summarized qualitatively as follows. Conventional cellular sensing with local processing has essentially no sensing-related fronthaul burden, but its localization accuracy is typically limited to several meters and its detection reliability is constrained by single-site geometry. Coordinated small-cell sensing modestly improves both metrics, but still lacks full network coherence. Centralized cell-free ISAC offers the best localization and detection performance, with representative localization accuracy in the sub-meter regime and detection probability approaching unity in favorable scenarios, but it may require aggregate fronthaul on the order of several Gbps and incurs significant fusion latency. Fully distributed cell-free architectures greatly reduce fronthaul requirements and improve scalability, but they generally deliver somewhat lower sensing performance and may require multiple consensus iterations to converge. The key lesson is that no single processing architecture dominates across all metrics.

This observation has direct system implications. Applications such as autonomous driving and safety-critical robotics, which demand high localization accuracy and low miss probability, are more naturally aligned with centralized or selectively centralized cell-free processing, provided that fronthaul is sufficiently provisioned. Industrial sensing and indoor automation often favor hierarchical architectures, which balance latency, complexity, and performance. Large-area environmental monitoring and persistent surveillance, by contrast, are better matched to distributed processing, where scalability and coverage dominate over strict latency requirements. In this sense, fronthaul and the processing architecture define the feasible operating point of networked ISAC: they determine how much of the theoretical gain of distributed sensing can be translated into practical performance.

The transition from single-site sensing to networked ISAC should not be understood as an increase in the number of APs. It is a shift toward a new inference architecture in which representation, transport, and computation are tightly coupled. The central design problem is not simply how to distribute sensing, but how to distribute it without destroying the information needed for coherent fusion and timely decision-making. This systems perspective sets the stage for the next two issues: synchronization, which determines whether distributed observations can be phase-aligned, and interference management, which determines whether cooperation remains beneficial at the network scale.

C. SYNCHRONIZATION AND COOPERATIVE SENSING LIMITS

The performance gains of networked and multi-static ISAC depend not only on having many distributed observations, but also on accurately aligning them for joint processing. In principle, cooperative sensing benefits from geometric diversity, broader spatial coverage, and multi-perspective illumination of targets. In practice, however, these gains are realized only if signals collected at different APs can be synchronized in time and phase with sufficient precision. Synchronization is therefore a foundational systems constraint in distributed ISAC: it determines whether spatial diversity can be converted into coherent gain, or whether the network is forced to operate in a weaker non-coherent regime.

This issue is especially severe because the synchronization requirement is set not by communication timing alone, but by the much stricter needs of coherent sensing. Consider two APs observing the same target. If the received signals are to be combined coherently, the residual timing mismatch

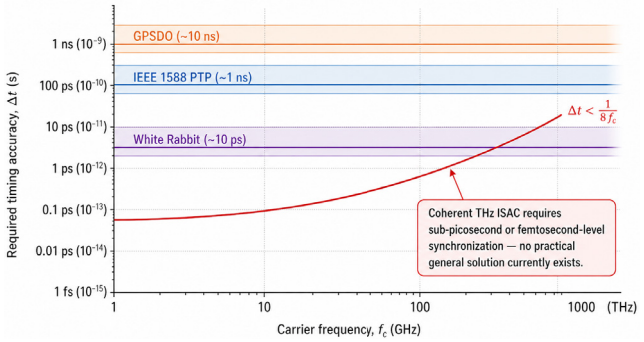


FIGURE 9. Illustration of the synchronization gap in coherent multi-static ISAC. The required timing accuracy for coherent sensing scales inversely with carrier frequency, making sub-6 GHz cooperation challenging and coherent THz cooperation beyond the reach of conventional network synchronization methods.

Δt between the two APs induces not only a range bias, but also a carrier-phase error and a distortion of Doppler estimation. These effects may be summarized qualitatively as follows: timing error shifts the inferred propagation delay, leading to target position bias; phase misalignment destroys coherent summation and reduces sensing SNR; and timing inconsistency across coherent processing intervals perturbs Doppler extraction. Among these effects, phase coherence is typically the most restrictive. Requiring the phase error to remain below $\pi/4$ so that coherent gain loss stays within approximately 3 dB yields the simple condition

$$\Delta t < \frac{1}{8f_c}, \tag{7}$$

where f_c is the carrier frequency. Equation (7) immediately reveals the severity of the problem: the timing budget shrinks inversely with carrier frequency. At 3.5 GHz, the allowable error is on the order of tens of picoseconds; at 28 GHz, only a few picoseconds; and at THz frequencies, the requirement approaches the sub-picosecond or even femtosecond regime. This is far beyond what conventional cellular synchronization mechanisms were designed to deliver. Figure 9 visualizes this synchronization gap by comparing coherent-sensing timing requirements across carrier frequencies.

The origin of this gap is not a single error source, but the accumulation of several distinct impairments. A useful system-level decomposition is

$$\Delta t_{\text{total}} = \Delta t_{\text{clock}} + \Delta t_{\text{prop}} + \Delta t_{\text{ADC}} + \Delta t_{\text{proc}}, \tag{8}$$

where the terms represent clock offset and drift, propagation asymmetry in the synchronization channel, sampling-clock jitter, and processing-latency variation, respectively. Clock drift arises when each AP oscillator deviates slightly from its nominal frequency, leading to phase offsets that accumulate over time even after initial calibration. Propagation asymmetry appears because practical synchronization protocols often assume symmetric delay, whereas real fronthaul links and hardware paths rarely satisfy that assumption exactly. ADC jitter introduces irreducible random timing noise at the sampling stage, while processing latency variation reflects load-dependent delay in digital pipelines. Importantly, these components do not simply add in a benign way: at high carrier frequencies, even small residuals in any one term can dominate the total coherence budget.

The impact of synchronization error on distributed sensing can be quantified. For a coherent multi-static system with K APs, let the residual alignment error after synchronization and delay compensation at AP k be ϵ_k. If the envelope distortion is neglected and the impairment is phase rotation, the normalized coherent gain can be written as

$$G_{\text{coh}} = \frac{\left|\sum_{k=1}^{K} \alpha_k e^{-j2\pi f_c \epsilon_k}\right|^2}{\left|\sum_{k=1}^{K} \alpha_k\right|^2}, \tag{9}$$

where α_k denotes the complex amplitude of the return from AP k. For equal amplitudes and independent Gaussian timing errors $\epsilon_k \sim \mathcal{N}(0, \sigma_\epsilon^2)$, where σ_ϵ^2 denotes the per-AP residual timing-error variance, the expected coherent gain becomes

$$\mathbb{E}[G_{\text{coh}}] = \frac{1}{K}\left(1 + (K-1)e^{-4\pi^2 f_c^2 \sigma_\epsilon^2}\right). \tag{10}$$

This follows because each cross-term contains $\mathbb{E}\left[e^{-j2\pi f_c(\epsilon_k - \epsilon_\ell)}\right]$ and $\text{Var}(\epsilon_k - \epsilon_\ell) = 2\sigma_\epsilon^2$ for independent per-AP timing errors. Equivalently, if $\sigma_{\Delta\epsilon}^2$ is defined as the pairwise timing-error variance, the exponential term can be written as $e^{-2\pi^2 f_c^2 \sigma_{\Delta\epsilon}^2}$. Accordingly, Equation (10) should be interpreted as a reference coherent-combining behavior under the stated equal-amplitude, independent-Gaussian-error, and phase-only assumptions, rather than as a deployment-independent performance guarantee; practical residual timing errors may also be correlated across access points and shaped by the underlying clock-distribution architecture.

This expression exposes a sharp operating threshold. When σ_ϵ is sufficiently small, the network benefits from coherent summation, and the effective sensing precision improves rapidly with AP cooperation. Once σ_ϵ exceeds a critical value, however, the exponential term collapses, and the gain approaches that of incoherent averaging. In other words, adding more APs beyond that threshold provides little benefit unless synchronization is improved simultaneously. This transition is one of the most important practical limits in distributed ISAC. In the coherence-limited regime, localization accuracy can scale approximately with both bandwidth and the number of coherently combined APs, whereas in the incoherent regime, the gain from cooperation is much weaker and may be dominated by geometry rather than phase alignment. As a result, synchronization should not be viewed merely as an implementation nuisance; it is a structural determinant of whether cell-free ISAC operates as a true coherent sensor network or as a loosely fused collection of local detectors.

Current synchronization technologies illustrate this gap clearly. GPS-disciplined oscillators are attractive because they are simple and widely available, but their residual timing

error is typically in the nanosecond range under realistic conditions, which is inadequate for coherent multi-static sensing. IEEE 1588 Precision Time Protocol over fiber improves performance substantially, but still struggles to guarantee the picosecond-level stability needed beyond sub-6 GHz operation. White Rabbit offers the best demonstrated performance among practical networked timing systems, achieving sub-nanosecond and, in controlled settings, tens-of-picoseconds accuracy, but this requires dedicated infrastructure, careful calibration, and highly stable links. Two-way over-the-air time transfer, backhaul-based carrier-phase synchronization, and distributed MIMO clock-alignment methods remain promising research directions, but none currently provides a general solution for large-scale coherent ISAC under realistic wireless deployment conditions. The implication is particularly stark at THz frequencies: coherent THz ISAC is limited not only by propagation and hardware nonlinearities, but by a synchronization requirement that lies beyond the reach of conventional electronic clock-distribution methods.

For system design, the key conclusion is that synchronization constrains architecture as strongly as fronthaul or computation. If synchronization accuracy is insufficient, the network should not be modeled or advertised as a fully coherent cell-free sensor, because the predicted gains from joint processing will not materialize in practice. Instead, the processing strategy must be adapted to the achievable timing regime, for example, by restricting coherent fusion to tightly synchronized clusters while relying on non-coherent track fusion across clusters. This observation also directly couples with the architecture discussion in the previous subsection: the effective processing topology of distributed ISAC is determined jointly by fronthaul and synchronization, not by either factor alone. More broadly, synchronization defines the boundary of realism for cooperative sensing. A network with excellent geometry but poor timing may underperform a smaller but tightly synchronized deployment. Conversely, progress in picosecond-level synchronization could unlock a qualitatively different sensing regime in which large numbers of APs contribute coherent multi-static gain at mmWave and possibly sub-THz frequencies. This makes synchronization one of the central bottlenecks in the roadmap toward scalable, high-resolution networked ISAC.

D. INTERFERENCE MANAGEMENT AT NETWORK SCALE

As ISAC evolves from isolated links to dense multi-cell and cell-free deployments, interference ceases to be a secondary impairment and becomes a defining system-level constraint. Unlike conventional communication networks, where interference is viewed as an additive degradation in the signal-to-interference-plus-noise ratio, networked ISAC must contend with a broader, more structured interference landscape. In particular, interference may degrade data transmission, corrupt sensing observations, create ghost targets, mask genuine targets, or distort clutter estimates. Interference management in networked ISAC is not merely a resource-allocation problem; it is also a sensing-integrity problem.

A useful distinction is between communication-oriented and sensing-oriented interference. The former includes familiar communication-to-communication interference between simultaneous links, as well as sensing-to-communication interference when high-power sensing transmissions leak into downlink or uplink reception. The latter is more distinctive to ISAC. Communication-to-sensing interference arises when communication signals contaminate sensing receivers, often in a structured rather than purely Gaussian manner because the interfering waveform may be known but misaligned in time or space. Even more problematic is sensing-to-sensing interference, especially in distributed or multi-cell settings, where one node's sensing transmission may be received as a direct-path leakage, a cross-echo, or a clutter-mediated reflection at another node. Such terms can generate false alarms or ghost targets rather than merely reducing detection SNR, making them qualitatively different from conventional interference in communication theory.

This distinction is particularly important in cooperative multi-static sensing. Consider two neighboring APs that both perform sensing in the same band. A return received at one AP may originate not only from its own transmission, but also from another AP's probing waveform after reflection from a common target or from the surrounding environment. If these components are not modeled and canceled correctly, the receiver may interpret them as additional targets or as target displacement. Thus, dense sensing cooperation creates a paradox: the same increase in spatial diversity that improves localization accuracy also increases the number of unintended sensing paths that must be managed.

At network scale, these effects are naturally studied through stochastic geometry and large-system analysis, which make clear that densification improves sensing through geometry and diversity only up to the point at which synchronization limits, fronthaul burden, and structured sensing interference begin to dominate [55]. If AP locations are modeled as a spatial point process with density λ_{AP}, then increasing AP density improves communication coverage and shortens average link distance, but it also increases the aggregate sensing interference seen by each receiver. The resulting sensing signal-to-interference ratio is therefore not monotone in deployment density. In sparse networks, additional APs improve sensing through diversity and geometry. In dense networks, however, uncoordinated or weakly coordinated sensing transmissions may produce an interference floor that offsets the benefit of additional nodes. This implies the existence of an operating regime in which densification is beneficial, and another in which further densification yields diminishing or even negative returns on sensing.

Several mitigation strategies follow from this observation. Orthogonalization in time is the simplest option, allowing neighboring APs to take turns transmitting sensing waveforms. This can suppress cross-sensing interference, but

at the cost of reduced temporal resolution and a lower network-wide sensing duty cycle. Frequency partitioning offers a more flexible alternative, but fragments bandwidth and can reduce both communication spectral efficiency and sensing resolution. Code-domain separation and waveform diversity may further reduce structured interference, provided that cross-correlation properties are well controlled. More ambitious approaches rely on cooperation: APs exchange waveform information, timing offsets, or local estimates to predict and cancel known interference. However, such strategies inherit the synchronization and fronthaul burdens discussed in the previous subsections, and their effectiveness deteriorates when timing or phase knowledge is inaccurate.

The key systems insight is that interference management in ISAC cannot be designed independently of architecture. Centralized and tightly coordinated networks can, in principle, suppress or explain much of the structured interference that arises in sensing, because multiple waveform sources and propagation paths are explicitly modeled. Distributed networks, by contrast, benefit from lower transport cost and greater scalability, but often lack the global information required for precise interference cancellation. This creates a three-way coupling among interference, fronthaul, and synchronization: interference cancellation requires shared information, shared information consumes fronthaul, and accurate cancellation is only possible if the relevant nodes are sufficiently synchronized. Accordingly, interference should be treated as a first-order scaling constraint in distributed ISAC. The relevant question is not simply how to suppress interference locally, but how to maintain a net positive cooperation gain as network density increases. This perspective motivates the final subsection, which synthesizes architecture, fronthaul, synchronization, and interference into a unified view of system-level scaling laws and tradeoffs.

E. SYSTEM-LEVEL SCALING LAWS AND TRADEOFFS

The preceding subsections highlight a central fact: networked ISAC performance is not determined by any single component in isolation. Architecture, fronthaul, synchronization, and interference are tightly coupled, and the practical value of distributed cooperation depends on how these factors scale jointly with network size. A system-level perspective is therefore essential. The relevant question is no longer whether cooperation is beneficial in principle, but under what scaling regime the gains from additional APs exceed the penalties introduced by transport overhead, synchronization error, processing latency, and mutual interference.

The first and most important scaling variable is AP density. Increasing the number of APs generally improves geometric diversity, shortens propagation distance, and creates more transmitter–receiver pairs for multi-static sensing. In the ideal coherent regime, these effects can translate into substantial improvements in detection probability and localization accuracy. However, the gain is not unbounded. As AP density increases, fronthaul load rises, synchronization across larger distributed infrastructures becomes more difficult, and cross-sensing interference becomes more severe. The resulting performance curve is therefore typically non-monotone: initial densification improves sensing quality, but beyond a certain point, the marginal benefit of additional APs is eroded by the system's overhead needed to coordinate them.

A second key tradeoff is between coherent and non-coherent cooperation. Coherent fusion offers the highest theoretical gain because it combines both amplitude and phase information across APs, enabling sharp localization and strong weak-target detectability. Yet it requires the most stringent timing accuracy and the largest fronthaul burden, particularly when raw or lightly compressed observations must be transported for centralized fusion. Non-coherent cooperation is much more scalable, since APs may exchange detections, tracks, or local estimates rather than raw waveforms, but it gives up a substantial portion of the achievable sensing gain. In practice, coherent cooperation tends to dominate in small or moderate networks with strong infrastructure support, whereas non-coherent or partially coherent cooperation becomes increasingly attractive in large-area deployments.

Latency introduces a third systems tradeoff. Centralized architectures can deliver the best sensing accuracy, but only after aggregating and jointly processing large amounts of distributed data. This aggregation delay may be acceptable for environmental monitoring or infrastructure sensing, but it can be problematic for high-mobility applications such as autonomous driving or robotic control. Distributed inference reduces transport latency and avoids a single fusion bottleneck, but often requires iterative consensus or message passing, which may itself introduce multiple rounds of delay before convergence. Thus, low-latency ISAC is not guaranteed by distributed processing alone; it depends on whether the corresponding distributed algorithm converges faster than a centralized architecture can aggregate and fuse the data.

These coupled constraints imply that no single architecture dominates all performance metrics. Broadly speaking, conventional cellular sensing offers low infrastructure complexity and minimal sensing-specific transport cost, but limited localization accuracy and weak geometric diversity. Small-cell coordination modestly improves these metrics, but does not fully exploit the multi-static structure. Centralized cell-free ISAC provides the best sensing performance when fronthaul and synchronization are sufficient, often enabling sub-meter localization and near-unity detection probability in favorable scenarios, but at the cost of high transport load and substantial processing concentration. Hierarchical architectures offer a compromise by capturing strong local cooperation while limiting global overhead. Fully distributed architectures provide the best scalability and are attractive for very large deployments, but they generally deliver weaker sensing performance and depend heavily on algorithmic convergence properties. In this sense, system design in networked ISAC is best understood as selecting an operating

point on a multi-dimensional Pareto frontier rather than searching for a universally optimal solution.

Application requirements determine where that operating point should lie. Safety-critical and mobility-sensitive applications favor architectures that prioritize localization accuracy and low miss probability, even if this requires higher fronthaul expenditure or stronger infrastructure support. Industrial automation often benefits from hierarchical cooperation, in which local coherence can be exploited without global fusion across the entire network. Wide-area monitoring and persistent surveillance are more naturally matched to distributed architectures, where coverage and scalability dominate over strict latency or sub-meter positioning. The design of networked ISAC should therefore be application-driven: the optimal balance among AP density, fronthaul, synchronization, and interference suppression depends on whether the task prioritizes accuracy, coverage, latency, energy efficiency, or deployment cost.

Networked ISAC also enables sensing-driven network control. Instead of using sensing only as an output service, the network can feed environmental state estimates back into control loops. For example, sensed user trajectories can support AP selection and handover preparation; blockage maps can trigger multi-connectivity or proactive beam switching; crowd-density estimates can support load balancing and scheduling; and sensed interference or clutter maps can guide spatial reuse and waveform coordination. In cell-free deployments, distributed sensing can also inform which APs should participate in coherent fusion, which should serve communication users, and which should remain idle to reduce interference or energy consumption. The central challenge is that sensing outputs are uncertain, delayed, and sometimes privacy-sensitive, so network-control policies must account for confidence, latency, and data exposure.

The broader lesson is that distributed ISAC should not be evaluated solely on isolated link-level gains. A design that appears highly favorable at the waveform or geometry level may become unattractive once fronthaul, synchronization, and interference scaling are taken into account. Conversely, a seemingly modest architecture may become the superior choice once system overhead is accounted for. This observation reinforces the central message of this section: the future of networked ISAC depends not only on stronger physical-layer designs but also on a systems theory that explains when cooperation scales, saturates, and becomes counterproductive.

V. EMERGING REGIMES AND ENABLING TECHNOLOGIES

While the preceding sections focused on the core foundations and system architectures of ISAC, several emerging regimes are now reshaping the assumptions underlying most existing designs. In particular, large-aperture arrays, extremely high carrier frequencies, programmable electromagnetic environments, and data-driven transceiver design are pushing ISAC into operating regions where conventional far-field, narrowband, and modular-processing abstractions become increasingly inadequate. These developments are not merely incremental enhancements; they introduce qualitatively new propagation behaviors, hardware constraints, and algorithmic opportunities. This section examines the principal emerging regimes and enabling technologies that are expected to define next-generation ISAC, with emphasis on how they alter the underlying sensing–communication trade space and motivate new models, new signal processing methods, and new deployment strategies. Federated sensing, discussed in Section VII-D, further complements these enabling technologies by supporting distributed intelligence without centralized raw sensing-data collection.

A. NEAR-FIELD ISAC

Most existing ISAC formulations assume far-field propagation, under which the impinging wavefront is approximately planar, and angle and range can be treated as separable quantities. This approximation is valid only when the target distance r satisfies the classical Fraunhofer condition

$$r \geq \frac{2D^2}{\lambda}, \tag{11}$$

where D is the effective array aperture and λ is the carrier wavelength. As array apertures increase and carrier frequencies move toward the mmWave and THz regimes, this condition is often violated in practical deployments. In large-aperture RISs, holographic MIMO platforms, and dense short-range sensing scenarios, targets often lie in the Fresnel region, where spherical wavefronts dominate and conventional far-field approximations no longer apply. Near-field ISAC should therefore be understood not as a minor correction to existing models, but as a distinct operating regime of growing practical importance.

The essential difference is that, in the near field, spatial phase no longer depends only on direction. Instead, the phase profile across the array carries coupled information about both angle and range, so that two targets located at the same angular direction but at different distances may produce distinguishable wavefront curvature. This is a fundamental departure from far-field beamsteering, where targets sharing the same angle are difficult to separate unless additional delay diversity, wideband processing, or multiple snapshots are available. In the near field, by contrast, a sufficiently large aperture can exploit spherical-wave structure to resolve range directly from spatial measurements. The result is a qualitatively new sensing capability: single-shot three-dimensional localization using phase curvature across the array.

For a linear array with element position d_m measured along the array axis and angle θ measured with respect to the array broadside, the propagation distance to a target at polar coordinates (r, θ) is

$$\ell_m(r, \theta) = \sqrt{r^2 + d_m^2 - 2\, r d_m \sin\theta}, \tag{12}$$

rather than the linear far-field approximation used under planar-wave assumptions. The corresponding phase progression

is nonlinear across the aperture, and this nonlinearity encodes range information in addition to direction. In ISAC, this coupling is not a nuisance; it is a new sensing resource. It enables the array to distinguish same-angle targets at different ranges, improves range–angle resolution without relying on bandwidth expansion, and creates opportunities for more precise joint localization and tracking. Figure 10 illustrates how near-field wavefront curvature enables range-aware discrimination and beamfocusing at specific angle–range coordinates.

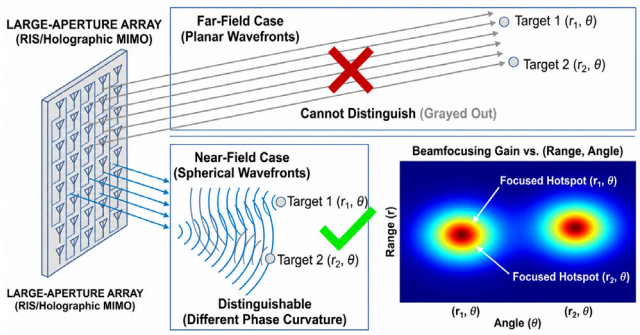


FIGURE 10. **Near-field ISAC beamfocusing schematic. Under far-field planar-wave propagation, two targets located at the same angular direction but different ranges are difficult to distinguish using spatial processing alone. Under near-field spherical-wave propagation, the curvature of the received phase profile varies with range, enabling range-aware discrimination and beamfocusing at specific angle–range coordinates.**

This change in propagation geometry also changes the meaning of beamforming. In the far field, beamforming primarily steers energy toward a direction. In the near field, beamforming becomes beamfocusing: the array can concentrate energy at a specific point in space characterized jointly by angle and range. Such a point of focus is attractive for both sensing and communications. On the sensing side, it allows the system to interrogate a selected range–angle region while suppressing clutter and interference outside the focal zone. On the communication side, it enables users located at similar angles but different distances to be spatially separated, creating a form of range-aware spatial multiplexing with no direct far-field counterpart. More broadly, near-field beamfocusing suggests that future ISAC systems may be designed not merely to illuminate directions, but to sculpt localized electromagnetic hotspots in space.

A useful physical interpretation is that large-aperture arrays convert geometry itself into a sensing resource. Once the propagation regime enters the Fresnel zone, the traditional separation between beamforming and localization begins to blur: the array response becomes inherently range-sensitive, and localization performance depends not only on bandwidth and time delay, but also on aperture size and spatial phase curvature. This is especially relevant in short-range 6G scenarios such as indoor robotics, industrial automation, vehicular safety sensing, RIS-assisted links, and THz sensing, where large apertures and short wavelengths make near-field operation common rather than exceptional. For example, with a 0.5 m aperture operating at 28 GHz, the near-field region extends to approximately 50 m according to Equation (11), which already covers a large fraction of indoor and dense small-cell use cases.

These observations have immediate consequences for channel modeling and transceiver design. Existing ISAC channel models, including those adapted from conventional communication standards, are predominantly far-field and therefore unable to capture spherical-wave propagation, distance-dependent focusing, and mixed near- and far-field scattering. A near-field ISAC channel model must account for at least three features. First, the propagation geometry must retain explicit wavefront curvature rather than collapsing each path to an angle-only representation. Second, beam patterns must be described in terms of angle–range focusing behavior rather than direction-only steering gain. Third, the model must support hybrid regimes in which some paths are near-field while others remain well approximated by far-field propagation within the same scene. These requirements are particularly important for RIS-assisted and THz ISAC, where small wavelengths and large effective apertures make near-field behavior unavoidable.

From an estimation perspective, near-field ISAC also motivates new performance benchmarks. Classical CRB analyses based on far-field assumptions may significantly underestimate the achievable gain of large-aperture arrays when range information is embedded directly in the spatial phase curvature. Conversely, transceiver architectures designed exclusively for far-field beamsteering may fail to exploit the full resolution benefit available in the Fresnel region. This suggests that near-field processing should not be viewed merely as a refinement of conventional beamforming, but as a new design axis that links aperture size, carrier frequency, localization accuracy, and sensing selectivity.

Overall, near-field ISAC represents one of the clearest examples of how emerging 6G hardware trends alter the sensing–communication trade space. What was treated as a breakdown of a convenient approximation becomes, in this regime, a source of additional structure that can be exploited for high-resolution localization, range-aware multiplexing, and controlled spatial interrogation. The challenge is that these gains are available only if the corresponding channel models, beamfocusing strategies, and estimation methods are redesigned to operate natively in the near field rather than treating spherical-wave effects as a small perturbation. This near-field perspective naturally connects to RIS-enabled ISAC, because RISs combine large effective apertures with programmable wave manipulation.

B. RIS-ENABLED ISAC

RISs have emerged as one of the most important enabling technologies for next-generation ISAC because they transform the radio environment from a passive propagation medium into a programmable sensing-and-communication

resource [56], [57]. In conventional communication settings, the primary purpose of an RIS is to create favorable propagation paths in non-line-of-sight links. In ISAC, however, its role is broader. A RIS can simultaneously reshape communication channels, redirect sensing beams into occluded regions, create virtual multi-static geometries, and enlarge the effective sensing aperture without requiring an active transceiver array [18], [56], [58]. In this sense, RIS-enabled ISAC is not merely communication with reflective assistance; it is a form of environment-programmable perception.

A useful starting point is the equivalent channel representation. For communication, the composite channel between transmitter and receiver may be written as

$$\mathbf{H}_{\text{tot}} = \mathbf{H}_{\text{direct}} + \mathbf{H}_{\text{rx,RIS}} \boldsymbol{\Phi} \mathbf{H}_{\text{RIS,tx}}, \tag{13}$$

where

$$\boldsymbol{\Phi} = \text{diag}\left(e^{j\phi_1}, e^{j\phi_2}, \ldots, e^{j\phi_M}\right) \tag{14}$$

is the RIS phase-control matrix with M elements. In ISAC, the sensing channel is richer because the RIS may participate not only in the forward communication path, but also in the round-trip sensing path through direct, RIS-assisted, reverse-RIS, and even double-RIS components. This creates a programmable sensing geometry in which reflected energy can be steered, focused, or coded through the surface configuration itself. The central implication is that RIS control changes not only the received signal strength, but the very observability structure of the scene.

From a systems perspective, RIS adds two forms of value. First, it enhances communication by creating virtual line-of-sight paths around blockages and by increasing effective beamforming gain for users located in difficult propagation regions. Second, and more distinctively for ISAC, it enhances sensing by serving as a controllable reflector or virtual aperture. Through appropriate phase programming, a RIS can redirect probing energy into blind zones, create additional bistatic paths, enable synthetic-aperture-like operation without mechanical motion, and improve angular selectivity in cluttered environments. These capabilities are attractive in indoor automation, urban canyons, and infrastructure sensing, where line-of-sight is often intermittent, and the environment itself becomes part of the sensing architecture.

RIS-enabled ISAC is not a single architecture, but a family of related designs with different operating tradeoffs. A *passive RIS* provides reflection-only control with low power consumption and limited hardware complexity, but suffers insertion loss and typically offers only phase control. An *active RIS* augments the surface with amplification, thereby boosting weak echoes and extending coverage, at the cost of increased power consumption, hardware complexity, and self-noise. *STAR-RIS* architectures further allow simultaneous transmission and reflection, enabling dual-sided coverage and more flexible sensing geometries. The common feature across these variants is that the RIS moves part of the transceiver intelligence into the propagation environment itself, but the practical design tradeoff differs sharply across passive, active, and dual-mode surfaces. Passive RIS is appealing when low cost and simplicity dominate; active RIS is attractive when echo strength is the bottleneck; STAR-RIS is most valuable when full-space coverage is required.

The achievable gain from RIS-assisted sensing depends strongly on the reference normalization. In an ideal phase-aligned passive RIS, the reflected field contributions from M elements can add coherently, so the received power term may scale as M^2 relative to a single reflecting element under fixed geometry and ideal lossless reflection. However, when the comparison is made relative to an incoherent M-element aperture, the incremental coherent-combining advantage scales as M, corresponding to $10\log_{10}(M)$ dB. For $M = 256$, this coherent-over-incoherent reference is approximately 24 dB, whereas the single-element ideal M^2 reference would be 48 dB. Both are reference upper-bound scaling behaviors under their respective normalizations and exclude two-hop path loss, insertion loss, phase quantization, mutual coupling, control latency, and hardware non-idealities. Therefore, the practical sensing gain should be interpreted as an end-to-end geometry- and hardware-dependent improvement rather than as a direct realization of the ideal scaling law. Figure 11 summarizes this distinction by comparing ideal coherent scaling with more practical passive, active, and STAR-RIS gain trends.

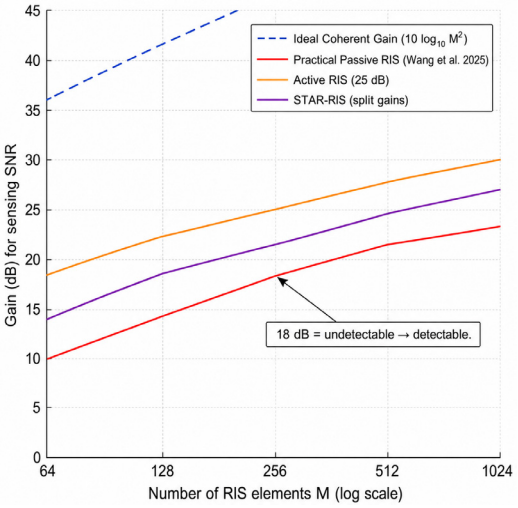


FIGURE 11. Representative RIS-assisted sensing gain versus the number of RIS elements. The ideal curve illustrates coherent combining behavior under a specified normalization, whereas practical passive, active, and STAR-RIS implementations achieve lower end-to-end gains due to two-hop path loss, phase quantization, mutual coupling, control latency, insertion loss, and hardware non-idealities. The curves are conceptual and should not be interpreted as deployment-independent link-budget predictions.

Beyond raw gain, RIS also enables sensing modes that are difficult to realize with conventional static propagation environments. A RIS can act as a virtual aperture for angle estimation, steer probing energy into occluded regions,

generate controllable multi-static paths, and support synthetic aperture or coded-reflection sensing modes. These sensing functions matter because they change the type of scene information available to the receiver, not merely its SNR. In particular, by introducing programmable reflection paths, RIS-assisted ISAC can create geometries in which targets that were previously weakly observable become strongly observable from a different angle or with a different clutter profile. This is especially attractive for non-line-of-sight target detection and indoor positioning, where geometry is often more limiting than transmit power. These benefits do not come for free. The hidden system cost of RIS-enabled ISAC is the control overhead. If a RIS with M elements uses b bits of phase resolution and must update its configuration at rate f_{update}, then the control signaling burden scales as

$$R_{\text{control}} = Mbf_{\text{update}}. \tag{15}$$

This overhead may remain modest in slowly varying scenes, but it becomes substantial when target mobility is high or when fast reconfiguration is needed for beam sweeping, tracking, or dynamic blockage mitigation. For large surfaces with fine quantization, the control channel itself can become a system bottleneck, especially in passive deployments that lack a dedicated high-rate backhaul. Thus, RIS control must be designed jointly with sensing objectives: updating too slowly reduces adaptivity, whereas updating too aggressively may consume more signaling resources than the gains justify.

RIS-enabled ISAC also exposes a major modeling gap. Existing channel models rarely account for RIS elements as programmable scatterers with realistic radiation patterns, inter-element coupling, phase quantization errors, and control latency. Yet these factors directly determine both communication gain and sensing fidelity. A usable RIS-aware ISAC channel model should therefore include not only the nominal RIS phase matrix, but also element-level non-idealities, bistatic reflection behavior, and the latency and granularity of configuration updates. Without such a model, simulation-based evaluations risk overstating the benefits of RIS assistance and understating the cost of practical implementation. This modeling requirement links RIS-enabled ISAC directly to the broader challenge of ISAC-native channel modeling discussed earlier in the paper.

A particularly important connection arises between RIS and near-field ISAC. Because RIS deployments often involve large apertures and relatively short target distances, many practical RIS-assisted sensing scenarios naturally fall in the Fresnel region rather than the far field. In this regime, the RIS no longer performs simple far-field beamsteering; instead, it can achieve beam focusing toward a specific angle–range point. The corresponding phase design must align the cascaded path from transmitter to RIS to target, and the resulting focusing gain depends explicitly on target range rather than only angle. This makes RIS a natural hardware platform for near-field ISAC and reinforces the view that large intelligent surfaces are not merely coverage-enhancement devices, but spatially programmable sensors. The combined near-field and RIS perspective is particularly relevant for dense indoor deployments and high-frequency systems, where the distinction between communications infrastructure and sensing infrastructure becomes increasingly blurred.

RIS-enabled ISAC should be viewed as a controlled reconfiguration of the propagation environment in service of joint communication and sensing objectives. Its promise lies not only in higher gain, but in altered geometry, improved observability, and environment-aware beam control. Its challenge lies in realistic modeling, control overhead, and hardware non-idealities. These features make RIS a compelling, but also highly system-dependent, enabling technology in the emerging ISAC landscape.

C. MILLIMETER-WAVE AND TERAHERTZ ISAC

mmWave and THz bands are among the most consequential spectral regimes for next-generation ISAC because they simultaneously expand communication bandwidth and sensing resolution [11], [12], [59]. At mmWave frequencies, multi-gigahertz channels already enable high-rate links and fine range resolution beyond what is feasible at sub-6 GHz. Moving further into the THz regime intensifies this trend: the enormous available bandwidth supports sub-centimeter and, in some windows, even millimeter-scale range resolution, while the high carrier frequency improves sensitivity to micro-motion and fine structural dynamics [12]. These features make mmWave and THz ISAC especially attractive for short-range high-resolution applications. At the same time, however, the very mechanisms that create this resolution also make the propagation channel more fragile, hardware more constrained, and joint sensing–communication design substantially more difficult.

At a high level, the attraction of these bands is straightforward. The classical range-resolution relation

$$\Delta R = \frac{c}{2B}, \tag{16}$$

shows that a larger bandwidth B directly improves sensing resolution. Thus, while a conventional cellular band offers meter-scale resolution unless a very wide aggregated bandwidth is used, mmWave and THz systems can naturally achieve centimeter- and sub-centimeter-scale resolution. At automotive radar frequencies around 77 GHz, bandwidth on the order of 1 GHz already yields range resolution of roughly 15 cm. At THz frequencies, bandwidths of 10–50 GHz can, in principle, reduce the resolution to the centimeter or even millimeter scale. This places THz ISAC in a qualitatively different sensing regime, where the system can resolve very fine surface structure, micro-Doppler signatures, and small-scale geometric variation that would be effectively invisible at lower carrier frequencies.

The challenge is that THz propagation is fundamentally unlike conventional microwave propagation. Molecular absorption, primarily from water vapor and oxygen, introduces strong frequency-selective attenuation, fragmenting

the band into a set of transmission windows rather than a single, uniformly usable spectrum. This means that wideband communication and wideband sensing do not automatically prefer the same part of the spectrum. A sensing function may desire the largest available aggregate bandwidth, even if that bandwidth is spectrally discontinuous, whereas a communication function may prefer a cleaner contiguous transmission window with lower attenuation and more stable signal quality. This molecular-absorption-induced fragmentation is a uniquely THz design constraint and is one of the main reasons why THz ISAC should be treated as a distinct regime rather than as a simple extension of mmWave ISAC.

A second challenge is hardware fragility. Oscillator phase noise grows with carrier frequency, power amplifiers deliver far less output power than in sub-6 GHz systems, and highly directional operation becomes mandatory to compensate for severe path loss. These impairments affect both sensing and communication. Increased phase noise degrades the communication error-vector magnitude, blurs Doppler estimation, and reduces coherent integration gain for sensing. Limited output power shortens communication range and weakens radar echoes, particularly in non-line-of-sight environments. As a result, THz ISAC is not a high-bandwidth version of conventional ISAC; it is an operating regime in which the link budget, waveform design, and signal-processing architecture are all tightly constrained by hardware non-idealities.

A third challenge is beam squint. At mmWave and THz frequencies, the relative bandwidth B/f_c can become large enough that the array response is no longer approximately frequency-flat. Different frequency components are then steered toward different spatial directions, producing distinct effective channels across the band. For communication, this reduces beamforming efficiency and complicates wideband precoding. For sensing, this creates a subtler problem: the full-band probing waveform and the communication sub-band may exhibit different spatial responses, so the sensing and communication functions no longer observe the same array geometry. This couples spectral allocation, beamforming, and estimation in a way that does not arise in narrowband systems. In ISAC terms, beam squint is not just a hardware nuisance; it is a source of sensing–communication mismatch that must be jointly accounted for in transceiver design.

These observations motivate a more nuanced view of THz opportunities. No single THz window optimizes all ISAC metrics simultaneously. Lower windows around the 0.14 THz range offer the best compromise among absorption loss, practical communication range, hardware maturity, and still-useful sensing resolution, making them the most plausible early candidates for deployable THz ISAC. Mid-band windows provide higher resolution but shorter range and more severe hardware penalties. Upper windows can enable extraordinary range resolution and micro-motion sensitivity, but only over very short distances and under tightly controlled environments. This tradeoff is summarized in Figure 12, which highlights the distinct operating points associated with the major THz transmission windows. In particular, the figure shows that the 0.14 THz window provides the best overall balance for early practical deployment, whereas upper windows sacrifice robustness and range in exchange for extreme sensing resolution.

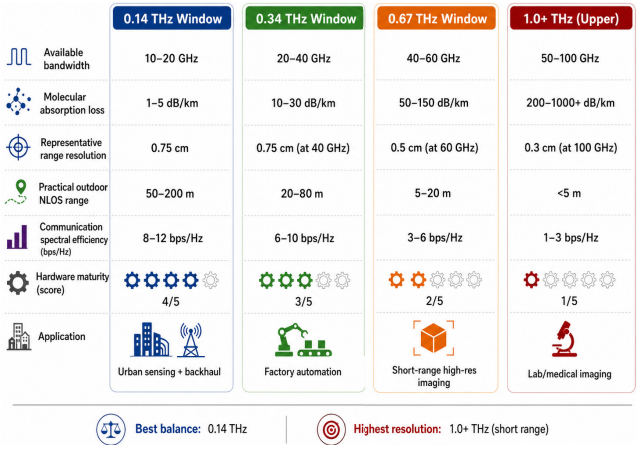


FIGURE 12. Representative tradeoffs across major THz transmission windows for ISAC. Lower windows offer a more favorable balance among bandwidth, absorption loss, practical range, and hardware maturity, while higher windows provide finer sensing resolution at the cost of rapidly degraded link robustness and stronger hardware constraints.

From a system-design perspective, mmWave and THz ISAC therefore create a new kind of Pareto problem. Higher carrier frequencies improve spatial and range resolution but also increase phase noise, reduce link robustness, and tighten alignment requirements. Wider bandwidth improves sensing fidelity, but may worsen beam squint and force communication to occupy spectrally cleaner sub-bands. Larger arrays compensate for path loss, but also increase calibration complexity and, in many cases, push the system toward near-field operation. These interactions make mmWave and THz ISAC deeply cross-coupled across propagation, hardware, and signal processing, reinforcing the need for designs that depart from conventional narrowband, far-field assumptions.

For this reason, THz ISAC should be viewed as a distinct emerging regime with its own core research questions. The central challenge is not merely to port existing mmWave beamforming or radar methods upward in frequency, but to characterize the achievable sensing–communication frontier under band fragmentation, hardware fragility, and frequency-dependent array response. In particular, a realistic THz ISAC design must decide how to partition spectral resources between communication and sensing objectives, how to jointly compensate for phase noise and beam squint, and how to exploit available transmission windows without assuming that all bandwidth is equally useful. These features make THz ISAC one of the most promising yet technically delicate frontiers in the broader ISAC landscape.

D. AI-ENABLED AND AI-NATIVE ISAC

AI and machine learning are rapidly becoming integral to ISAC because many of the underlying design problems are high-dimensional, non-convex, mismatched to the model, and strongly coupled across sensing and communication objectives [14], [60]. Existing work has already demonstrated the utility of data-driven methods for adaptive waveform design, beam management, resource allocation, channel synthesis, and online policy optimization [13], [61]. In most of these studies, however, machine learning serves as an enhancement layer on top of a conventional transceiver. This distinction is important. In current ISAC practice, learning is primarily AI-enabled or AI-assisted: it improves selected modules within a modular physical layer, but does not fundamentally redefine the end-to-end architecture. By contrast, the more ambitious 6G vision is AI-native ISAC, in which the entire transceiver is learned jointly as a single mapping from raw observations to both communication and sensing outputs.

This difference is more than semantic. In an AI-assisted design, the transceiver remains decomposed into interpretable blocks such as channel estimation, equalization, demodulation, decoding, beamforming, and resource allocation, while learning is used to replace or optimize individual modules. Such designs are attractive because they retain the structure and verification logic of classical communication systems while still benefiting from data-driven adaptation. For ISAC, this can be useful when the channel is partially unknown, the environment is difficult to model analytically, or the control policy must adapt online under strict latency constraints. In this regime, learning supplements model-based design rather than replacing it. Figure 13 contrasts this AI-assisted use of learning with the more ambitious AI-native transceiver model. This trend is also consistent with work on autonomic and self-reconfigurable wireless transceivers, where online signal analytics and software-defined adaptation are used to improve link robustness and reduce reliance on fixed hand-crafted operating points [62].

AI-native ISAC goes much further. Instead of preserving a hand-designed modular pipeline, it treats the full transceiver as a trainable input–output system. Raw IQ samples, or related low-level measurements, are fed into a neural architecture, such as a transformer or convolutional network, which directly produces decoded bits alongside sensing outputs, such as range, Doppler, angle, or target state. The transceiver is optimized end-to-end using a joint loss that reflects both communication and sensing objectives, for example, a weighted combination of bit-level cross-entropy and sensing error or estimation-theoretic criteria. In principle, this allows the network to discover latent representations that serve both tasks simultaneously, rather than imposing a fixed separation between communication processing and sensing processing. This is precisely where AI-native ISAC becomes most compelling: its architecture is naturally aligned with ISAC's multi-objective nature.

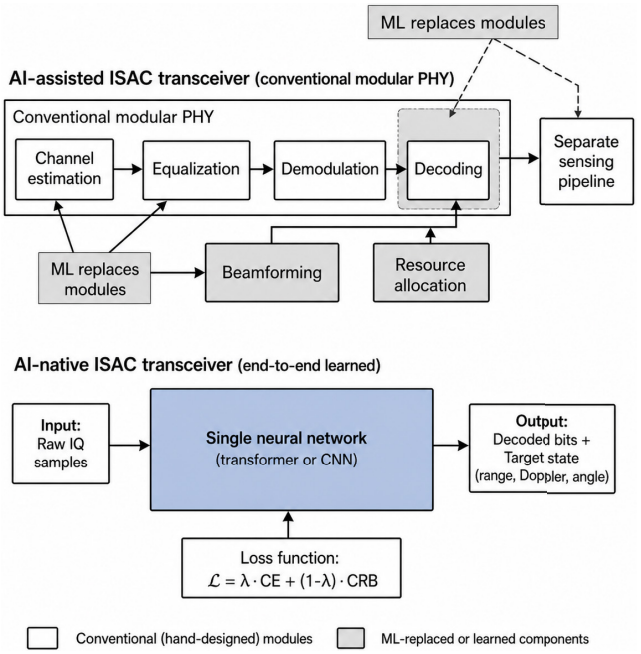


FIGURE 13. Distinction between AI-assisted and AI-native ISAC transceivers. In AI-assisted architectures, machine learning replaces or optimizes selected modules within a conventional communication and sensing pipeline. In AI-native architectures, a single learned model maps raw observations to decoded bits and sensing estimates through joint end-to-end training.

Different AI model families are likely to play distinct roles within ISAC. Graph neural networks are natural for cell-free and RIS-assisted ISAC because APs, users, targets, and RIS elements form graph-structured interaction networks. Transformers are useful for temporal context modeling, beam tracking, mobility prediction, and multimodal fusion because they can represent long-range dependencies across time, frequency, and sensors. Reinforcement learning is well-suited to adaptive waveform selection, scheduling, beam management, and resource allocation, in which actions affect future sensing and communication states. Generative models can support channel synthesis, rare-event simulation, digital-twin augmentation, and data generation for environments where measurements are scarce. Foundation models may eventually support semantic scene interpretation and cross-modal reasoning, but they introduce major challenges in latency, energy consumption, robustness, interpretability, and certification. Thus, AI-native ISAC should not be understood as a single neural architecture, but as a family of model choices whose suitability depends on the task, timescale, data availability, and safety requirements.

There are several reasons why ISAC is particularly well-suited to this AI-native perspective. First, ISAC is inherently multi-task. A conventional design must explicitly balance communication rate, sensing accuracy, detection performance, and resource usage through carefully engineered tradeoffs. A learned end-to-end architecture, by contrast, can optimize a joint objective directly and may discover

internal feature representations that are useful for both tasks simultaneously. Second, ISAC channels are often only partially observable in a model-based sense. Sensing measurements contain side information about targets, scatterers, motion, and blockage that may be difficult to encode using closed-form algorithms but can nonetheless be exploited by a learned transceiver. Third, many hardware impairments that complicate analytical design—including phase noise, nonlinearity, and IQ imbalance—may be easier to compensate through in-situ learning than through explicit parametric modeling. These features make AI-native ISAC more than a fashionable add-on; they make it a plausible architectural response to the fundamental complexity of joint sensing and communication.

At the same time, AI-native ISAC introduces a profound gap between theory and practice. Classical transceiver design admits explicit performance analysis, interpretability, and certification relative to known objectives such as capacity, error probability, or CRBs. A fully learned transceiver typically does not. Even if it performs well empirically, it remains unclear how close it is to the underlying information-theoretic capacity–distortion frontier, whether its behavior is robust under distribution shifts, and how its decisions can be certified in safety-critical settings. The challenge is not merely to demonstrate that learned transceivers can outperform heuristic baselines, but to establish under what conditions they can approach fundamental limits in a provable and reliable manner. Without such guarantees, AI-native ISAC will remain difficult to standardize and difficult to trust in high-consequence deployments.

This concern is especially important because the transition from AI-assisted to AI-native ISAC is not incremental. Replacing a beam-selection block or a resource allocator with a neural network does not fundamentally alter the system's verification framework. Replacing the entire sensing and communication pipeline with a single learned mapping does. Once the transceiver becomes end-to-end learned, questions of explainability, calibration, robustness, repeatability, and performance certification become first-order design constraints rather than downstream implementation details. In this sense, AI-native ISAC is as much a problem in systems assurance and theoretical characterization as it is a problem in machine learning. The near-term deployment pathway is therefore likely to be asymmetric. AI-assisted ISAC is already plausible across several application domains because it can be integrated into existing architectures with minimal structural disruption. AI-native ISAC is more likely to emerge first in constrained environments where channel statistics are relatively stationary, abundant training data are available, and the task scope is narrow enough to keep model complexity manageable. Representative examples include indoor factory automation, limited-range industrial sensing, wake-on-motion systems, and testbed environments supported by digital twins. In these settings, it may be realistic to train and validate a learned transceiver against a well-understood baseline and to evaluate whether the learned design closes the gap to jointly optimized model-based systems.

Overall, AI-enabled and AI-native ISAC should be viewed as two different stages in the evolution of learning for 6G systems. The first stage uses machine learning to improve selected components within a conventional transceiver; the second rethinks the transceiver itself as a learned multi-task inference engine. The former is already technically useful and increasingly practical. The latter is far more transformative, but also far less mature, because it depends on a new theory for certification, limit characterization, and trustworthy deployment. This is why AI-native ISAC occupies a distinctive place in the emerging ISAC landscape: it offers one of the boldest opportunities for architectural simplification and joint optimization, yet it also exposes one of the deepest unresolved gaps between empirical machine learning success and communication-theoretic rigor. Digital twins, discussed in Section V-F, provide a complementary mechanism for AI-native ISAC by offering synchronized simulation environments for training, validation, and closed-loop policy testing before deployment.

E. ENERGY-EFFICIENT ISAC DESIGN

Energy efficiency is a central design objective for 6G systems, and this requirement is particularly important for ISAC, as joint sensing and communication must often be sustained continuously at high bandwidth under stringent latency and reliability constraints. In principle, ISAC offers an immediate energy-efficiency advantage over separately deployed sensing and communication systems: by sharing spectrum, waveform generation, RF front-ends, antenna arrays, and parts of the baseband processing chain, it can reduce duplicated hardware and lower static power consumption. ISAC is not only a spectral-efficiency concept, but also a hardware-sharing concept with potentially significant energy benefits.

The difficulty is that these savings are not automatic. The sensing function introduces its own power costs, and in many cases, these costs grow precisely in the operating regimes where ISAC becomes most attractive. For example, radar-style probing and target detection often favor high peak power, long coherent integration, or persistent illumination, all of which can conflict with communication-oriented energy minimization. Similarly, full-duplex or simultaneous transmit–receive operation, while appealing for monostatic sensing, requires self-interference cancellation chains and high-linearity front-end components whose power consumption can offset part of the hardware-sharing gain. Thus, energy-efficient ISAC is not achieved simply by merging two systems into one; it requires a joint design that explicitly accounts for both dynamic and static power expenditures.

A major source of inefficiency lies in the waveform and amplifier operating point. Many promising ISAC waveforms, including OFDM- and OTFS-based designs, exhibit relatively high PAPR. High PAPR forces the power amplifier to operate with output backoff to preserve linearity, thereby

directly reducing drain efficiency. This problem is especially acute in wideband and high-frequency systems, where power amplifiers are already less efficient and more sensitive to nonlinearity. From an ISAC perspective, the problem is doubly important: waveform distortion degrades not only communication reliability, but also sensing fidelity, range sidelobe structure, and Doppler estimation quality. Energy-efficient ISAC, therefore, requires waveform design that considers not only spectral efficiency and sensing resolution, but also the amplifier efficiency implied by the transmit signal statistics.

Another important design axis is power allocation across communication and sensing objectives. In conventional communication systems, transmit power is typically controlled to achieve a target rate or reliability subject to an energy budget. In ISAC, the same transmitted energy also determines target detectability, parameter estimation accuracy, and clutter robustness. A system optimized only for communication may underilluminate the scene, while one optimized only for sensing may overconsume energy for communication. The resulting design problem is inherently multi-objective: transmit power, beam shape, bandwidth usage, duty cycle, and hardware operating point must be selected jointly to achieve an acceptable balance between communication throughput, sensing performance, and total energy consumption.

This coupling becomes even stronger at the hardware level. The total energy cost of ISAC is not determined solely by radiated power, but by the full transceiver chain, including data converters, local oscillators, digital signal processing, beamforming networks, and any auxiliary circuits required for synchronization, control, or self-interference cancellation. At mmWave and THz frequencies, this burden becomes even more severe because high-speed converters, phase-noise mitigation, and large-array calibration can consume a large share of the overall energy budget. In distributed or cell-free architectures, energy consumption must also include fronthaul transport and network-side processing. Accordingly, energy-efficient ISAC should be understood as a cross-layer design problem spanning waveform design, hardware architecture, signal processing, and network coordination.

A useful way to frame the problem is through energy efficiency as a joint utility, rather than a communication-only metric. Instead of measuring efficiency solely in bits/Joule, ISAC invites broader notions, such as the joint rate-and-sensing utility per Joule, or weighted combinations of throughput, localization accuracy, and detection performance, normalized by total power consumption. Such formulations are important because they expose tradeoffs that would otherwise remain hidden. For example, a modest increase in hardware power may be justified if it yields a large reduction in sensing error or allows communication and sensing to share infrastructure that would otherwise be duplicated. Conversely, a nominally high-rate design may be energetically unattractive if it relies on inefficient waveforms, excessive control overhead, or persistently active sensing modes.

The practical implication is that energy-efficient ISAC cannot be understood as low-power communication with an added sensing function. Rather, it requires the co-design of waveform selection, transmit power, beamforming strategy, hardware operating point, and task-level utility. Architectures that appear optimal in terms of communication rate or sensing accuracy alone may become unattractive once amplifier efficiency, static circuit power, and network-side overhead are incorporated into the evaluation. Energy efficiency, therefore, becomes a natural cross-cutting theme across the full ISAC stack and directly connects to the broader problem of hardware- and deployment-aware ISAC design. For this reason, energy-efficient ISAC should be regarded not as a secondary implementation concern, but as a necessary condition for the practical deployment of large-scale 6G sensing-and-communication systems. More fundamentally, this perspective underscores the need for a hardware-aware ISAC theory, since the value of joint sensing and communication depends not only on achievable rate and estimation performance, but also on how efficiently these gains can be realized in real transceiver architectures.

F. DIGITAL TWIN-ASSISTED ISAC

Digital twins are synchronized virtual representations of physical systems and provide a natural architectural complement to ISAC. The relationship is bidirectional: ISAC supplies real-time environmental observations that keep the twin synchronized, while the twin provides a controlled environment for algorithm validation, AI-native training, what-if analysis, and closed-loop control. In this role, a digital twin is not merely a visualization layer; it becomes part of the sensing–communication control loop.

Recent work on digital twin-assisted ISAC has shown how a digital twin can reconstruct a belief state from delayed telemetry and support latency-robust reinforcement learning for joint beamforming and power allocation [63]. This is especially relevant for cloud-RAN and networked ISAC deployments, where sensing decisions may be based on delayed or incomplete network observations. Related work on network and edge digital twins has also shown how DTs can support VNF migration, vehicular-network optimization, scalable autonomous-vehicle twins, and programmable edge-resource provisioning [64], [65], [66], [67]. These capabilities are complementary to ISAC-DT because they provide the management, simulation, and control substrate needed for deployment-grade perceptive networks. Digital twins can also support deployment calibration by comparing predicted sensing performance against measured data, stress-testing AI-native policies under rare events, and quantifying uncertainty before field operation.

Open challenges include synchronization fidelity between the physical environment and the twin, scalable modeling of dense urban or industrial scenes, uncertainty quantification,

sim-to-real transfer for learned policies, and twin security as an attack surface. Standardization work on network digital twins and management-plane automation is relevant, but dedicated digital twin-ISAC standards remain immature. For this reason, digital twin-assisted ISAC should be viewed as an enabling architecture for deployment-grade validation, AI-native training, and closed-loop control rather than as a standalone sensing modality.

G. LOW-POWER ISAC: LPWAN AND BACKSCATTER SENSING

Most ISAC research emphasizes high-rate communication, high-resolution sensing, and infrastructure-class hardware. A complementary regime lies at the opposite extreme: low-power ISAC for massive IoT. In this setting, the goal is not centimeter-level radar imaging, but opportunistic sensing, zone-level localization, event detection, and multi-year battery life. LPWAN technologies such as LoRaWAN and NB-IoT are attractive because they offer long-range connectivity at low power, but their narrow bandwidth and low duty cycle make high-resolution radar-style sensing unrealistic.

Low-power ISAC should therefore be understood as opportunistic or semi-passive sensing. Instead of transmitting dedicated radar waveforms, the system reuses communication observables, such as received signal strength, time of arrival, channel-state variation, and packet-level statistics, as coarse sensing probes. Backscatter and wake-up-radio architectures push this idea further: low-power or battery-free devices can modulate and reflect ambient RF signals, while wake-up receivers allow nodes to remain in an ultra-low-power state until a relevant RF event occurs [68], [69], [70]. In this regime, the reflected signal is not only a communication carrier for sensor data, but also a sensing signature shaped by tag state, motion, orientation, and the surrounding propagation environment. These architectures suggest an energy-centric form of ISAC, where each microwatt must support both connectivity and perception.

The resulting research challenges differ sharply from those in high-performance ISAC. Narrow bandwidth limits range resolution, coherent synchronization is often impractical, unlicensed-band interference is severe, and sensing accuracy is typically measured in meters or zones rather than centimeters. Low-power ISAC is not a replacement for infrastructure-grade ISAC, but a complementary regime for ubiquitous, sustainable, and battery-constrained 6G IoT deployments [70].

VI. APPLICATIONS, PROTOTYPES, AND STANDARDIZATION OUTLOOK

The practical significance of ISAC is most evident in vertical domains where communication and sensing are operationally intertwined. In these settings, the value of ISAC does not arise merely from combining two functions within a common hardware platform, but from enabling new forms of situational awareness, reducing mutual interference between previously separate subsystems, and repurposing communication infrastructure as a sensing asset. At the same time, real-world deployment exposes a recurring tension: infrastructures designed primarily for communication are rarely optimized for sensing geometry, waveform structure, observability, or estimation fidelity. This section, therefore, reviews the major application domains, summarizes prototype systems that are beginning to validate ISAC under realistic hardware constraints, and concludes with a standardization outlook that connects current technical progress to likely deployment pathways.

A. AUTONOMOUS VEHICLES AND INTELLIGENT TRANSPORTATION

Autonomous vehicles and intelligent transportation systems represent one of the clearest near-term deployment paths for ISAC. In this setting, a vehicle-mounted or infrastructure-assisted ISAC platform can simultaneously support vehicle-to-everything (V2X) communication and high-resolution radar sensing for obstacle detection, lane keeping, pedestrian tracking, and cooperative awareness. This dual functionality is especially attractive because the two tasks are inherently interdependent: communication supports coordination and intent sharing, while sensing provides the local perception required for safe motion planning. These requirements make autonomous and connected vehicles one of the clearest near-term verticals for ISAC, since they depend simultaneously on low-latency V2X connectivity, cooperative perception, and high-integrity localization [7], [71].

ISAC offers at least two concrete advantages over separately deployed automotive radar and vehicular communication systems. First, it avoids mutual interference between co-located radar and V2X radios, which is becoming increasingly important as vehicles carry multiple sensing and connectivity subsystems operating in adjacent or overlapping bands. Second, it allows communication reference signals and pilot structures to be reused as sensing probes, thereby extracting sensing value from transmissions that would already be required for communication. In this sense, transportation ISAC is not merely a matter of hardware consolidation; it is a way to turn the communication infrastructure itself into a cooperative perception layer for driving. Cooperative vehicular positioning is especially relevant because communication-assisted sensing gains are often realized through shared geometry, neighbor information, and learned fusion across vehicles and infrastructure [71]. The challenge, however, is that vehicular environments are highly dynamic, cluttered, and safety-critical. Targets are mobile, occlusions are frequent, and latency constraints are severe. These factors make transportation a natural stress test for many of the core ISAC problems discussed earlier in the paper, including synchronization, interference management, waveform design, and trustworthy adaptation. As a result, autonomous driving is likely to remain one of the most

important verticals for determining whether ISAC can move from theoretical promise to deployment-ready capability.

B. SMART CITIES AND ENVIRONMENTAL MONITORING

Smart-city and environmental-monitoring scenarios extend ISAC from individual platforms to infrastructure-scale sensing. The central vision is that the communication network itself—base stations, APs, and distributed radio nodes—can simultaneously serve as a large-scale sensing system. Such an infrastructure could support traffic-flow estimation, crowd-density monitoring, weather sensing, structural health monitoring, and related forms of urban situational awareness without requiring a separate sensing network.

The appeal of this approach lies in scale and reuse. Cellular infrastructure is already geographically distributed, power-provisioned, network-connected, and continuously active. If even a portion of this infrastructure can be repurposed for sensing, then urban perception may be achieved at far lower deployment cost than would be required for dedicated radar installations. In effect, the city is covered not only by a communication fabric but also by a distributed, passive, or semi-active sensing fabric. This broader infrastructure perspective is also consistent with recent smart-city-oriented ISAC surveys, which emphasize traffic monitoring, crowd sensing, urban localization, and environment-aware infrastructure as natural deployment targets for networked sensing [20]. Industrial and smart-factory environments are particularly compelling for infrastructure-based ISAC because wireless connectivity, localization, safety monitoring, and coordinated robotic operation are tightly coupled in the same physical space [8]. At the same time, this vertical highlights one of the key limits of practical ISAC: communication infrastructure is not usually designed for sensing performance. Reference signal patterns, allocated bandwidth, array orientation, site placement, and inter-site geometry are optimized primarily for connectivity and coverage rather than target observability or localization accuracy. As a result, smart-city ISAC is fundamentally a repurposing problem, and its success depends on how effectively communication-oriented infrastructure can be adapted to sensing-oriented objectives. This makes city-scale monitoring a particularly relevant application context for cross-layer design, deployment-aware optimization, and standardization-aware ISAC.

C. HUMAN ACTIVITY RECOGNITION AND HEALTHCARE

Human activity recognition and healthcare illustrate a different but equally important dimension of ISAC: contactless, continuous sensing enabled by communication-capable infrastructure. Radar-based monitoring of respiration, heart rate, gesture dynamics, posture, and gross body motion has already been demonstrated in laboratory and clinical settings. ISAC extends this paradigm by enabling such monitoring functions to be integrated into the same wireless infrastructure that provides connectivity, thereby reducing the need for dedicated sensing hardware and making continuous observation more practical in homes, clinics, and assisted-living environments. Contactless vital-sign monitoring is especially representative of this convergence, since the same RF infrastructure can, in principle, support both wireless service and clinically relevant respiration or heartbeat estimation without requiring dedicated wearable instrumentation [72].

This vertical is particularly attractive because many medically relevant signals are weak, periodic, and local, making them well suited to high-resolution, short-range radio sensing. At the same time, healthcare-related ISACs impose constraints that differ somewhat from those in transportation or smart-city deployments. Reliability, false-alarm control, robustness to clutter and multipath, and interpretability of sensing outputs are all essential. In addition, privacy concerns are especially significant because the same infrastructure that enables non-contact monitoring may also infer sensitive behavioral or physiological information. For this reason, healthcare ISAC is not only an application opportunity, but also a compelling motivation for privacy-preserving and trustworthy ISAC design.

D. UAVs, AIRSPACE MANAGEMENT, AND NON-TERRESTRIAL NETWORKS

Uncrewed aerial vehicles (UAVs) and low-altitude airspace management provide another strong application driver for ISAC. The rapid proliferation of small UAVs in urban and suburban airspace has created a sensing demand that conventional air-traffic-control radar systems were never designed to meet. Small drones are difficult to detect, may fly at low altitude in cluttered environments, and often require persistent monitoring over populated areas. ISAC-enabled base-station networks are therefore attractive as a distributed aerial-surveillance infrastructure capable of detecting, tracking, and potentially identifying UAVs while continuing to serve terrestrial communication users. This use case has already emerged as one of the most visible near-term ISAC verticals because aerial targets combine demanding detection geometry with strong dependence on communication-assisted coordination and tracking [73]. Figure 14 summarizes this space–air–ground view of NTN-ISAC, highlighting the heterogeneous links and deployment constraints that distinguish NTN-ISAC from purely terrestrial systems.

A related extension is non-terrestrial ISAC, where wide-area coverage, orbital dynamics, and intermittent visibility create a sensing–communication regime that differs from terrestrial infrastructure and may require distinct energy, geometry, and control abstractions [74]. NTN ISAC extends the ISAC concept from terrestrial infrastructure to low-Earth-orbit (LEO) satellites, high-altitude platform stations (HAPS), and integrated terrestrial–non-terrestrial networks. NTN-ISAC should be treated as more than a niche extension of UAV sensing, because NTN is becoming a central component of 6G architecture. Recent standards-facing work on intent-to-install for 6G NTN/SAG further suggests

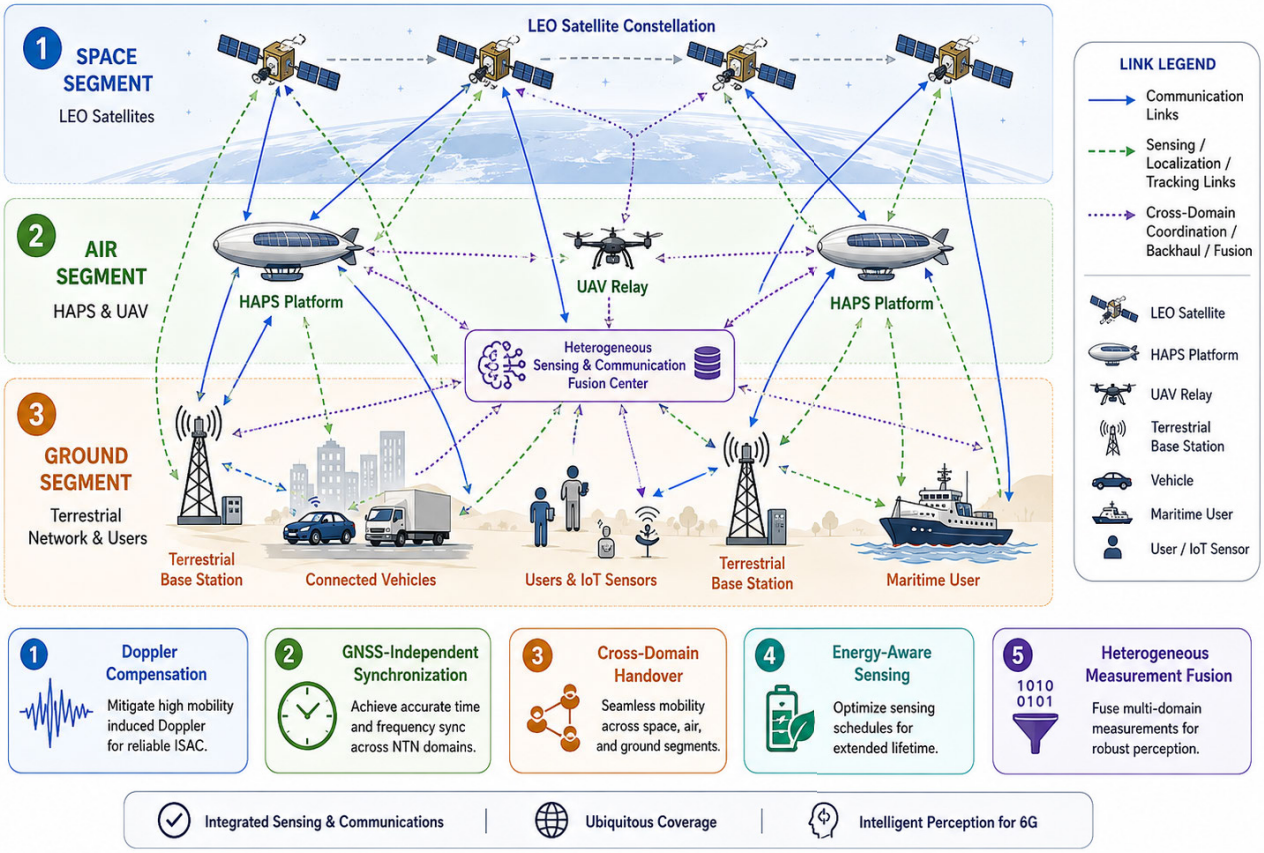


FIGURE 14. Space–air–ground integrated NTN-ISAC architecture for 6G. LEO satellites, HAPS, UAVs, and terrestrial base stations jointly support heterogeneous sensing and communication across the space, air, and ground segments, with key challenges including Doppler compensation, GNSS-independent synchronization, cross-domain handover, energy-aware sensing, and heterogeneous measurement fusion.

that future NTN deployments will require not only link-level optimization, but also intent-driven procedures for translating deployment objectives into installable, verifiable, and interoperable space–air–ground configurations [75]. On the communication side, LEO and HAPS systems must cope with high Doppler shifts, long or variable propagation delays, intermittent visibility, handover, and limited onboard energy. On the sensing side, their wide-area coverage and rapidly varying geometries create opportunities for environmental monitoring, localization, and distributed multi-static observation at scales beyond the reach of terrestrial infrastructure [76], [77], [78].

LEO satellite ISAC and HAPS-assisted ISAC expose different operating points. LEO constellations provide global coverage and rich bistatic geometries, but face severe Doppler, synchronization, and energy constraints [79]. HAPS operate closer to users, with lower latency and finer spatial resolution than satellites, while still covering much larger areas than terrestrial base stations [80]. These platforms make NTN-ISAC a natural extension of the networked ISAC issues discussed in Section IV: synchronization, fronthaul or inter-satellite transport, interference, and distributed fusion reappear at planetary scale. At the same time, NTN-ISAC raises sharper privacy and governance questions because wide-area sensing may cover populations and regions not explicitly associated with the serving network.

A full treatment of NTN-ISAC would require a separate survey. For this roadmap, the key point is that NTN-ISAC is an active, standards-relevant extension of networked ISAC. Near-term work should focus on Doppler-resilient waveforms, GNSS-independent synchronization, energy-aware sensing modes, heterogeneous terrestrial network–NTN fusion, and experimentally grounded validation. Ground-based platforms such as Starlink-enabled robotic testbeds can support early measurement studies, but in-orbit experimentation remains a longer-term bottleneck [81].

E. EXPERIMENTAL PROTOTYPE SYSTEMS

Experimental validation has progressed beyond conceptual demonstrations and now serves as a reality check for ISAC theory. Recent work includes standardization-oriented channel-modeling measurements [27], prototype demonstrations of shared sensing–communication architectures [82], and learning-assisted beam-management studies showing that environment awareness can reduce communication overhead [61]. Together, these results show that a subset of ISAC concepts has moved from abstract tradeoff analysis

to reproducible, hardware-oriented evaluation. The indoor and outdoor software-defined radio (SDR) prototype figures included in this section (i.e., Figures 15-16) show representative unpublished laboratory platforms from the authors' laboratory. They are used to illustrate hardware-realistic distributed ISAC implementation issues—including synchronization, receiver geometry, obstacle interaction, and multi-node coordination—rather than to define standardized benchmark performance. Unless quantitative metrics are explicitly stated, the conclusions drawn from these figures are architectural and experimental-contextual rather than deployment-general.

Several results illustrate this progress. Shared-band dual-function radar–communication testbeds have demonstrated simultaneous target detection and Mbps-level communication using a single spectrum and hardware, thereby validating the co-design principle in controlled outdoor settings. Millimeter-wave prototypes have shown that sensing-derived angular information can reduce beam-training overhead compared with exhaustive search, confirming that sensing can improve communication procedures rather than merely coexist with them. Sub-6 GHz RIS-assisted prototypes have reported sizable sensing gains in occluded geometries, indicating that programmable environments can provide not only communication enhancement but also meaningful perception benefits. Field-trial activities associated with emerging standardization efforts have further shown that indoor ISAC-enabled positioning with sub-meter accuracy is plausible in structured deployment scenarios. Beyond these shared-band and infrastructure-oriented demonstrations, a recent trend is the emergence of SDR-based distributed prototypes that make synchronization, geometry, and hardware constraints explicit parts of the experiment. Figure 15 shows an indoor SDR-based prototype in which synchronized USRP radios, a shared timing reference, distributed receivers, and controlled obstacle interaction are incorporated into a single experimental platform. Such testbeds are useful because they expose practical implementation constraints—especially timing alignment, receiver geometry, and hardware coupling—that are often abstracted away in simulation-based ISAC studies.

Another direction is the use of outdoor distributed prototypes that emphasize sensing geometry. Figure 16 illustrates an SDR-based outdoor platform in which spatially separated receivers are synchronized through a shared timing reference and used for receiver-side localization of a transmitter. This type of testbed is valuable because it operationalizes several of the system-level issues discussed earlier in the paper, including multi-static geometry, distributed inference, synchronization, and the transition between sensing and communication roles. These laboratory setups do not constitute standardized benchmark experiments, but they illustrate the hardware, synchronization, geometry, and coordination issues that must be addressed in practical multi-node ISAC implementations. Importantly, they reveal the remaining gaps. Some challenges are still algorithmic, such as robust joint processing and trustworthy learned transceivers. Others are architectural, such as network synchronization, fronthaul-efficient cooperation, and deployment-calibrated channel modeling. More broadly, published prototype systems can serve as empirical filters on the research agenda by identifying which theoretical ideas remain robust under practical hardware and propagation constraints.

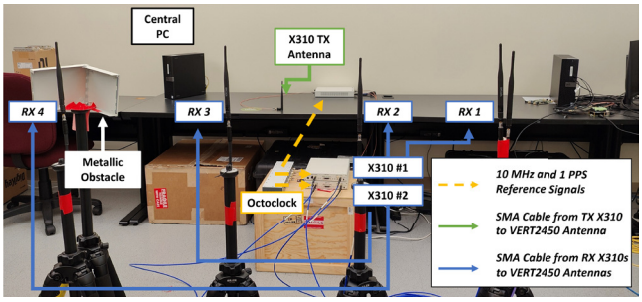


FIGURE 15. Representative unpublished indoor SDR laboratory setup for studying joint communication and obstacle-aware sensing. The platform uses synchronized USRP X310 radios, a shared Octoclock timing reference, distributed receivers, and a controlled metallic obstacle to examine timing alignment, distributed observation geometry, and hardware-realistic sensing–communication coupling. It is included solely as an architectural and experimental-context illustration, rather than as a standardized benchmark or deployment-general performance validation.

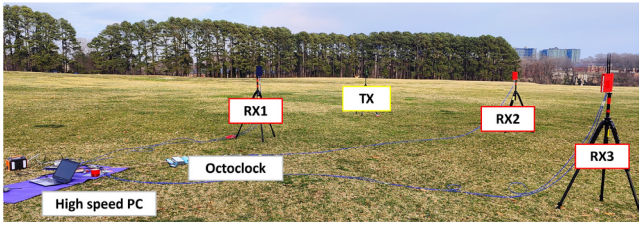


FIGURE 16. Representative unpublished outdoor SDR laboratory setup for studying distributed receiver-side transmitter localization. Spatially separated receivers are synchronized through a shared Octoclock timing reference and coordinated by a high-speed PC, providing a platform for examining distributed sensing geometry, timing consistency, and hardware-realistic localization in multi-node ISAC. It is included solely as an architectural and experimental-context illustration; waveform, bandwidth, measurement-protocol, and performance choices remain implementation-dependent, and no standardized benchmark claim is made.

F. STANDARDIZATION OUTLOOK

Standardization is the bridge between experimental plausibility and large-scale deployment, and ISAC has now progressed far enough that this bridge is becoming technically meaningful rather than aspirational. The key trend is that ISAC is entering standardization by first focusing on use-case definition, study-item framing, and deployment-oriented architecture discussions, rather than making an immediate hard commitment to a single waveform or transceiver structure. This is a sensible progression. The field is mature enough to justify system-level evaluation and service-level requirements, but many core design choices—particularly those involving channel modeling, synchronization, distributed

cooperation, and hardware-aware optimization—remain too open to premature freezing.

From a standards perspective, the most near-term path is through infrastructure-supported sensing functions that can be layered onto existing cellular systems with limited disruption. This includes positioning, environmental perception, UAV detection, and sensing-assisted communication procedures such as beam management. Such functions align with current cellular infrastructure, enable incremental deployment, and leverage capabilities introduced through reference-signal design, scheduling support, sensing interfaces, and architectural hooks rather than a redesign of the air interface. In this sense, early standardization is likely to favor ISAC functions that are deployment-compatible, infrastructure-centric, and closely tied to specific vertical use cases.

At the same time, standardization is likely to force the field to confront issues that are easier to postpone in academic work. One is interoperability: sensing outputs must be represented, exchanged, and fused consistently across vendors, infrastructure types, and deployment scenarios. Another is calibration and certification: communication systems can tolerate mismatch or implementation variability, whereas sensing systems are judged by localization error, detection reliability, and false-alarm behavior. A third is model realism. If ISAC is to be standardized for deployment, evaluation cannot rely on abstract Gaussian assumptions or communication-only channel models. This is why ISAC-native channel modeling, deployment-calibrated benchmarks, and experimentally grounded operating assumptions become central for standards-facing research.

A further challenge is that standardization is likely to proceed unevenly across application domains. Functions such as infrastructure sensing, positioning, and sensing-assisted beam management are plausible early candidates because they are aligned with current network architectures and offer clear operational value. By contrast, more ambitious directions—such as fully networked cell-free sensing, THz ISAC, or AI-native learned transceivers—are likely to remain outside the first wave of standardization because they depend on unresolved theoretical foundations, substantial architectural changes, or stronger certification requirements. This asymmetry should not be interpreted as a limitation of ISAC; rather, it reflects the field's progression from capability demonstration to deployable system design. Related work on network digital twins and NTN standardization is also relevant to ISAC deployment, since digital twins can support validation and management-plane automation, while NTN work defines the space–air–ground integration constraints under which future ISAC services may operate [76], [77].

The broader implication is that standardization will reward designs that are not only high-performing but also measurable, reproducible, and architecturally compatible with the evolution of existing networks. For the ISAC community, this means that performance gains alone are no longer sufficient. Results must be interpretable in terms of deployable interfaces, realistic channel assumptions, infrastructure constraints, and migration paths from current systems. In this sense, standardization does more than codify ISAC progress; it reshapes the research agenda by forcing the field to distinguish between what is theoretically interesting and what is practically incorporable into next-generation wireless systems.

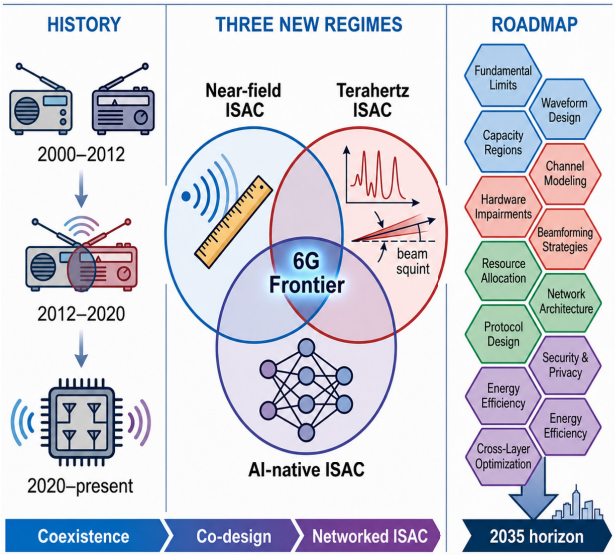


FIGURE 17. Conceptual overview of the ISAC research roadmap. The figure links the historical evolution of the field to three emerging regimes—near-field ISAC, THz ISAC, and AI-native ISAC—and organizes the main research challenges into a long-horizon roadmap for 6G systems.

VII. GRAND RESEARCH CHALLENGES

The preceding sections have shown that ISAC has progressed from a conceptual merger of sensing and communication into a broad research field spanning information theory, waveform and hardware design, distributed architectures, programmable environments, high-frequency operation, machine learning, and deployment-aware system design. Yet the field remains far from mature. Many open questions are not incremental optimization problems, but structural challenges whose answers will determine whether ISAC becomes an architectural foundation of 6G or remains a collection of fragmented techniques. In this section, we consolidate these unresolved issues into a coherent research agenda organized around four themes: foundational theory; modeling and physical-layer design; distributed and standards-oriented system design; and, finally, privacy, security, and cross-layer intelligence. Figure 17 summarizes this roadmap by linking the field's historical evolution, emerging regimes, and major challenge clusters.

A. FOUNDATIONAL THEORY CHALLENGES

At the deepest level, ISAC still lacks a complete theory of its joint limits in realistic multi-user settings. Existing information-theoretic results have clarified aspects of

single-link Gaussian models and a small number of special broadcast configurations, but these are only partial answers to the question that matters for large-scale deployments: what is the full capacity–distortion region when multiple users, multiple receivers, and multiple estimation objectives interact over the same wireless medium? As ISAC moves from isolated links to coordinated infrastructures, the problem is no longer whether sensing and communication can coexist, but rather how their resources can be jointly allocated across a networked multi-user system. A complete characterization of this region for canonical channels such as the multi-access channel, broadcast channel, and interference channel remains one of the clearest theoretical gaps in the field.

The difficulty is structural rather than merely technical. In a communication-only MAC, broadcast channel, or interference channel, the primary objects are message rates and decoding error probabilities. In ISAC, the same channel inputs also determine one or more environmental estimates, which may involve different targets, different receivers, and different sensing metrics. A transmission strategy may improve one user's communication rate while degrading another receiver's target observability; a waveform that improves detection probability may not minimize estimation error; and a sensing objective may be shared across users even when communication messages are private. These couplings make the capacity–distortion region multi-dimensional, metric-dependent, and sensitive to the topology of transmitters, receivers, and targets. Canonical models such as the ISAC MAC, ISAC broadcast channel, ISAC interference channel, relay-assisted ISAC, and cell-free ISAC should therefore be treated as priority problems for future theory.

A related limitation is that most current analyses remain asymptotic, assuming long blocklengths and vanishing error probabilities. These assumptions are poorly aligned with the short-packet, latency-sensitive operating conditions that motivate many practical ISAC applications, including vehicular safety, industrial automation, and edge autonomy. While finite-blocklength theory is well established for communication-only systems, an analogous non-asymptotic framework for joint sensing and communication remains lacking in general form. What is needed is a theory that characterizes the simultaneous penalties in communication reliability and sensing distortion as functions of blocklength, target error probability, and operating regime, ideally in forms analogous to the classical finite-blocklength converse and achievability machinery. Without such a framework, many theoretical ISAC tradeoffs remain difficult to interpret in the regimes where 6G systems will actually operate. These two foundational problems are tightly linked. The first asks what the ultimate joint sensing–communication limits are in multi-user environments; the second asks how those limits change once one abandons the unrealistic comfort of infinite blocklength. Together, they define the theoretical basis on which the rest of the field must ultimately rest.

Finite-blocklength ISAC is also more complicated than applying communication-only dispersion results to an added sensing constraint. The relevant non-asymptotic performance vector may include decoding error probability, missed-detection probability, false-alarm probability, estimation distortion, and decision latency. These quantities need not improve or degrade together. For example, a shorter packet may reduce control latency but weaken coherent integration; a more reliable communication code may consume symbols that would otherwise improve Doppler resolution; and a lower false-alarm threshold may increase detection probability while producing more sensing-induced control errors. A useful finite-blocklength theory for ISAC must therefore characterize joint reliability–detection–estimation tradeoffs rather than only communication decoding error.

B. MODELING AND PHYSICAL-LAYER CHALLENGES

Even if the fundamental limits were known, the field would still face a realism gap at the channel, hardware, and transceiver-architecture levels. A recurring theme throughout this paper is that existing communication-oriented channel models are not sufficient for ISAC. They do not represent target reflectivity, clutter structure, bistatic geometry, target dynamics, or the mixed near-/far-field behavior that becomes important in large-aperture and high-frequency systems. Conversely, sensing-oriented models often omit the mobility, multipath, and fading phenomena that affect communication performance. What is missing is a unified ISAC-native channel model, supported by measurement campaigns, that can be used not only for academic evaluation but also for deployment-oriented benchmarking and standardization.

This modeling gap is tied to a set of physical-layer design challenges that become especially severe once ISAC moves into mmWave and THz bands or attempts more aggressive transceiver architectures. At high carrier frequencies, waveform design cannot be separated from hardware impairments: phase noise, amplifier nonlinearity, IQ imbalance, converter limitations, and energy consumption affect communication and sensing performance simultaneously. Existing waveform designs often assume ideal front ends and discuss hardware losses only as implementation details. In reality, hardware impairment and energy efficiency are not secondary concerns; they are part of the core optimization problem. A realistic ISAC waveform theory must incorporate sensing quality, communication rate, hardware non-idealities, and energy efficiency in a single design framework.

A similar argument applies to full-duplex ISAC. Simultaneous transmission and reception in the same band is naturally attractive because it enables an embedded monostatic sensing mode within a communication system. However, the self-interference entering the receiver can be tens of decibels stronger than the desired echo, and existing full-duplex cancellation architectures were not developed with ISAC-specific waveform structure or sensing objectives in mind. This leaves a significant gap between the conceptual promise and the practical operation. The field still lacks

a clear understanding of the achievable cancellation limit when the canceller has access to the exact transmitted ISAC waveform, and more importantly, how close practical architectures can come to that limit without destroying sensing fidelity.

Machine learning introduces a physical-layer challenge, but of a different kind. AI-assisted ISAC has shown empirical value in beam management, waveform selection, and resource allocation. Yet the more ambitious step toward AI-native ISAC, where the transceiver itself is learned end-to-end, exposes an unresolved issue: the lack of theoretical guarantees. A learned transceiver may perform well in empirical tests, but it remains unclear how close it is to the communication–sensing frontier, how robust it is under distribution shifts, and how it should be certified for deployment. This is not only a question of training better models. The question is whether learned transceivers can be analyzed, bounded, and trusted with the rigor that classical communication theory demands. In this sense, AI-native ISAC stands at the intersection of machine learning, information theory, and systems assurance. Taken together, these issues show that the ISAC physical layer is still far from settled. Channel realism, hardware-aware waveform design, practical full-duplex operation, and learned transceivers with guarantees all remain open, and progress on these fronts will determine whether the elegant tradeoffs described in theory can survive contact with actual hardware and propagation environments.

C. DISTRIBUTED, NETWORK, AND STANDARDIZATION CHALLENGES

The next set of challenges arises when ISAC scales beyond a single node or a single link. In principle, distributed and cell-free architectures promise major gains through multi-static diversity, wider spatial coverage, and richer sensing geometry. In practice, however, these gains depend on coordination mechanisms that current wireless systems do not provide natively. One of the clearest bottlenecks is synchronization. Cooperative sensing requires timing and phase alignment that are far tighter than conventional communication systems were designed to support, especially when coherent fusion is desired across geographically distributed access points. Yet the effect of synchronization error on localization accuracy, coherent gain, and distributed sensing utility remains not as well characterized as communication theory provides for other forms of uncertainty. Without a better theory of synchronization-limited cooperation, it remains unclear when distributed ISAC is genuinely beneficial and when it collapses into little more than incoherent averaging.

Even with perfect synchronization, a broader network-level challenge remains: how to manage resources and interference when both sensing and communication must be coordinated across multiple cells, users, and targets. Unlike conventional communication interference, which is often treated as additional noise, sensing interference can be structured, spatially correlated, and capable of generating false targets or corrupted clutter estimates. This makes network-level ISAC resource management fundamentally more complex than the single-cell or communication-only counterpart. A satisfactory framework must jointly optimize spectrum use, power allocation, beam coordination, sensing duty cycle, and interference mitigation across the network, while accounting for the fact that communication and sensing interference have qualitatively different consequences.

These networked issues connect to the question of standardization and incremental deployment. ISAC is unlikely to be introduced through a clean-slate air interface. Instead, it will almost certainly have to emerge through partial reuse of existing reference-signal structures, frame formats, numerologies, deployment patterns, and infrastructure constraints inherited from 5G and evolving 6G systems. This means that standardization-friendly ISAC is not merely an implementation concern, but a research challenge in its own right. The field must determine how much sensing capability can be superimposed on existing communication infrastructure, what communication penalty this incurs, and what architectural migration path exists from communication-first deployments toward native ISAC systems. In other words, the problem is not simply to design a better ISAC waveform, but to design one that can survive the constraints of standards evolution and backward compatibility. Figure 18 places these challenges on an indicative maturity timeline, separating near-term enabling issues from longer-horizon theoretical and architectural problems.

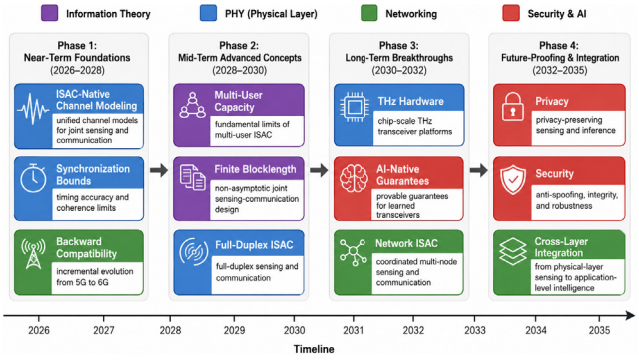


FIGURE 18. Research-agenda timeline for ISAC challenges. Different themes are likely to mature on different timescales, from near-term work on channel realism, synchronization, and deployment-compatible architectures to longer-term advances in AI-native, secure, and cross-layer ISAC.

This standardization momentum should also be interpreted in the context of the broader IMT-2030 framework, in which integrated sensing, context awareness, and intelligent network functionality are increasingly aligned with the long-term evolution of 6G capabilities [39], [46], [77]. These distributed and standards-facing challenges define the transition from technical feasibility to infrastructure-scale viability. They ask not only whether ISAC works, but whether it works in a form that can be coordinated, scaled, and incrementally deployed in real wireless systems.

D. PRIVACY, SECURITY, AND CROSS-LAYER INTELLIGENCE CHALLENGES

Privacy and security should be treated as first-class design dimensions in ISAC rather than as implementation concerns [22], [25]. Federated sensing, discussed below, is one possible mechanism for reducing raw sensing-data exposure while preserving collaborative learning across distributed infrastructure. Because ISAC-capable infrastructure can infer location, motion, posture, and even sensitive behavioral patterns, privacy risk arises not only from communication interception but also from environmental sensing. The privacy issue becomes more pronounced for NTN-ISAC, where LEO or HAPS platforms may provide wide-area sensing over regions and users that are not directly associated with a terrestrial base station. This makes authorization, purpose limitation, and sensing-access control essential research questions for future ISAC architectures. On the security side, adversarial attacks may target waveform manipulation, spoofed echoes, sensing-data poisoning, RIS-assisted attack surfaces, and joint communication–sensing control loops [21], [23], [24], [83], [84], [85]. These issues suggest that trustworthy ISAC requires joint treatment of physical-layer security, sensing privacy, and architecture-level resilience. Recent survey work makes clear that privacy in ISAC is not reducible to classical communication security alone, since the sensing function itself may reveal motion, occupancy, behavior, or environmental context even when payload confidentiality is preserved [23]. At the physical layer, the same probing structure that improves environmental observability may also expose communication-bearing signals, side information, or sensing signatures to malicious observers and targets [24].

Federated sensing. A promising approach to privacy-preserving ISAC is federated sensing, in which distributed ISAC nodes jointly train environment, beam-management, or resource-control models without exchanging raw observations. Each node uses local sensing and communication data to update a model, while only model parameters or gradients are shared with an aggregator or neighboring nodes [86], [87]. This idea is closely aligned with recent work on privacy-preserving edge learning and federated O-RAN control, where distributed learning is used to improve network intelligence while limiting raw-data exposure [88], [89], [90]. Applied to ISAC, such mechanisms could support shared occupancy models, classifier updates, beam-management policies, or resource-allocation rules without directly exposing location traces, channel matrices, user traffic, or sensed scenes.

Federated sensing does not eliminate the privacy and security problem. ISAC data are often highly non-IID across locations, rapidly time-varying, and coupled to mobility and blockage. Model exchange can also be expensive, and federated learning itself introduces new attack surfaces such as model poisoning and gradient leakage. Over-the-air aggregation and ISAC-aware edge learning may reduce communication cost, but they also couple learning performance to sensing noise, channel distortion, and synchronization error [91]. Federated sensing should therefore be treated as a research direction for trustworthy perceptual networks rather than as an already-solved privacy mechanism.

Security raises an equally important but broader concern. ISAC inherits all the familiar threats of wireless communication, including eavesdropping and jamming, but it also introduces sensing-specific attack surfaces. An adversary may inject false targets, manipulate clutter, exploit sensing observations to identify vulnerable communication nodes, or attack learned sensing modules through adversarial perturbations. These threats do not fit neatly into traditional communication security models because they involve both information leakage and integrity of perception. What is needed is a unified adversarial framework in which secrecy rate, false-alarm probability, sensing detection probability, and robustness to spoofing or jamming are treated jointly rather than as separate concerns.

Beyond privacy and security lies a more constructive challenge: cross-layer integration from physical-layer sensing to application-level intelligence. Physical-layer sensing produces delay, Doppler, angle, and related observables, but real applications care about richer abstractions such as object identity, intent, anomaly state, or action recommendation. ISAC cannot be fully evaluated at the waveform or estimation layer alone. It must also be understood as an inference pipeline in which physical-layer resource allocation affects edge inference, semantic representation, and ultimately decision quality at the application layer. This implies the need for a cross-layer ISAC design framework that links physical-layer control to application-level utility rather than treating sensing quality and downstream intelligence as independent stages. These final challenges broaden the field's scope in an essential way. They make clear that deployment-grade ISAC must be not only efficient and accurate, but also privacy-aware, secure, trustworthy, and meaningful at the application layer. In that sense, they define the transition from joint radio functionality to full-stack cyber-physical intelligence.

The grand research challenges described in this section are best understood not as a checklist of isolated open problems, but as a structured roadmap for ISAC's maturation. Some are near-term and enabling, such as realistic channel modeling, synchronization characterization, and deployment-compatible system design. Others are longer-horizon and transformative, such as AI-native guarantees and full cross-layer integration. Their common role is to identify the points at which current progress is most constrained by missing theory, missing models, or missing deployment realism. By articulating them in an integrated way, the objective is not to enumerate open questions, but to organize the field around the advances most likely to determine whether ISAC becomes a mature and enduring foundation of 6G systems.

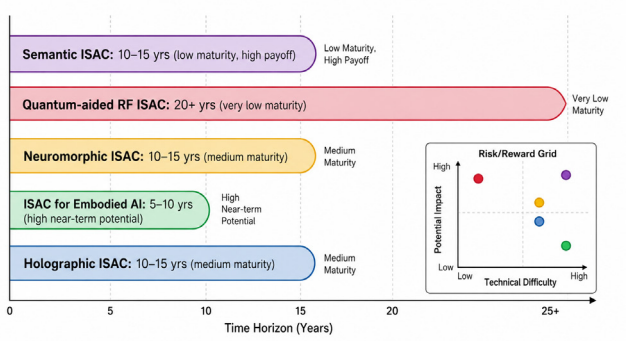


FIGURE 19. Horizon and risk–reward view of five transformative ISAC directions. The figure compares their maturity, time horizons, and payoffs, spanning from near-term embodied-AI integration to longer-horizon semantic, neuromorphic, holographic, and quantum-aided paradigms.

VIII. TRANSFORMATIVE RESEARCH DIRECTIONS

The grand research challenges discussed in the previous section are largely grounded in the near- to mid-term maturation of ISAC: they concern the theoretical, architectural, and deployment constraints that must be resolved if current ISAC concepts are to become scalable and standardizable 6G capabilities. Beyond these challenges, however, there is a second horizon of research directions whose significance is less immediate but potentially more transformative. They do not simply extend today's ISAC framework; they question its current abstractions and suggest that future sensing-and-communication systems may operate on very different foundations, including semantic information, quantum resources, event-driven computation, embodied decision loops, and nearly continuous electromagnetic apertures.

These directions are speculative but not arbitrary; each extends a limitation exposed by the grand-challenge roadmap. Bit-level communication may be too wasteful for task-driven systems; classical sensing may be far from the ultimate physical limit; conventional digital signal processing may be too energy-hungry for always-on edge perception; the separation of perception, communication, and action may be too slow for embodied autonomy; and discrete-array models may be too coarse for the spatial precision that future wireless environments demand. For these reasons, the directions discussed here should be understood not as short-term deliverables, but as plausible candidates for reshaping the field over the next decade and beyond. Figure 19 summarizes the relative maturity, time horizon, and risk–reward profile of the five directions discussed in this section.

A. SEMANTIC ISAC

A fundamental limitation of current ISAC is that it remains largely rooted in bit-level communication and parameter-level sensing. Communication is measured in bits per second; sensing is measured in terms of quantities such as detection probability, CRBs, or estimation error. Yet many practical applications do not require raw bits or full physical reconstructions; they require task-relevant meaning. Semantic ISAC extends this observation by asking whether communication and sensing can be aligned at the level of actionable information rather than solely on physical observables. In such a system, the radio would not attempt to communicate all available data nor sense every recoverable parameter, but would instead focus on those elements that matter for the downstream task, such as whether a pedestrian is present, whether a trajectory is anomalous, or whether a control action must be triggered.

The attraction of this direction is clear. If the relevant information is sparse relative to the raw sensed environment, then large efficiency gains may be possible in both communication and sensing. Rather than transporting detailed radar images, a system might communicate semantically compressed scene descriptions; rather than estimating all target parameters, it might prioritize only those needed for safety, planning, or inference. This could reduce bandwidth usage, processing cost, and latency in task-specific deployments. The challenge, however, is profound: semantic ISAC requires a theory that goes beyond Shannon-style source-channel separation and beyond conventional estimation theory, because the object of interest is no longer raw data fidelity but task relevance. Until a mature semantic information theory for joint sensing and communication exists, semantic ISAC will remain conceptually attractive but theoretically immature. It is best viewed as a long-horizon direction, with likely progress first in highly structured domains such as autonomous driving, industrial automation, or closed-world robotics, where the ontology of relevant objects and events is constrained.

B. QUANTUM-AIDED ISAC

If semantic ISAC challenges the informational abstraction of current systems, quantum-aided ISAC challenges their physical limit. The motivating idea is that quantum resources, such as entanglement, could, in principle, be used to simultaneously improve sensing and communication. Quantum illumination, for example, offers a theoretical advantage in detecting weakly reflecting targets in noisy environments, and more broadly the quantum metrology literature suggests that sensing performance may approach Heisenberg-limited scaling under appropriate conditions. A quantum-aided ISAC system would extend these ideas by asking whether the same quantum resource can support both sensing and communication functions within a unified architecture.

The potential payoff is extraordinary, but so are the obstacles. The gap between quantum optics demonstrations and practical RF wireless systems remains enormous. Quantum illumination has been explored experimentally in the microwave regime, but typically at cryogenic temperatures and over extremely short distances. Entanglement generation, distribution, storage, and preservation under room-temperature wireless RF propagation remain fundamentally unresolved problems. Moreover, the practical advantage of quantum sensing degrades rapidly in lossy channels, to the point where classical radar may outperform quantum illumination under most realistic wireless conditions. For

these reasons, quantum-aided ISAC should be treated with both seriousness and caution. It is genuinely transformative if made practical, but for RF wireless it remains a very long-horizon possibility rather than a near-term architectural candidate. Optical or hybrid free-space variants may mature sooner, but for mainstream wireless ISAC, this direction remains among the most speculative on the roadmap.

C. NEUROMORPHIC RADIO ARCHITECTURES FOR ISAC

A different path toward transformation comes from computation rather than physics. Current ISAC pipelines often rely on conventional digital processing architectures that are power-hungry and ill-suited to always-on perception in battery-constrained or edge-native systems. Neuromorphic computing offers a radically different model: event-driven processing using spiking neural networks, where computation is triggered by salient activity rather than repeated at every clock cycle. If such processors could be adapted for radio signal processing, they would enable ultra-low-power ISAC architectures in which sensing, inference, and communication processing are performed on a common neuromorphic substrate.

This direction is attractive because energy efficiency is likely to become a bottleneck for persistent ISAC in IoT nodes, mobile robots, and edge devices. A neuromorphic ISAC radio could, in principle, detect events, track sparse targets, and communicate only when relevant changes occur, thereby matching computation more closely to the temporal structure of the physical environment. The challenge is that ISAC signal processing remains different from many pattern-recognition tasks, where neuromorphic hardware has shown its clearest advantages. Radar and communication processing often require continuous-valued operations, a fine dynamic range, precise synchronization, and structured linear transforms that do not naturally map onto sparse event-based representations. As a result, neuromorphic ISAC is not merely about porting neural networks to a lower-power chip. It requires new algorithmic formulations that recast waveform processing, detection, and estimation in an event-driven computational paradigm. This suggests a promising direction for ultra-low-power edge ISAC, but remains medium- to long-term for general-purpose radio systems.

D. ISAC FOR EMBODIED AI

Among the transformative directions discussed here, ISAC for embodied AI is perhaps the most operationally grounded. Embodied agents—robots, autonomous vehicles, warehouse systems, and cyber-physical platforms—must continuously sense their environment, communicate with infrastructure or peer agents, and translate this information into timely action. Today, these functions are usually distributed across separate subsystems: sensing supports perception, communication supports coordination, and higher-level control determines action. ISAC proposes a different architecture in which sensing and communication are unified within a common perception–communication–action loop, so that the radio itself becomes part of the embodied intelligence pipeline.

The significance of this direction is not merely that one radio can do two jobs. Rather, the sensed environment can directly influence communication decisions, while communication can directly support collective perception and coordinated action. A robot may adapt beam management based on sensed geometry; a vehicle may use shared radio perception to anticipate occlusion; a factory agent may jointly optimize communication and sensing based on control urgency. This direction, therefore, operationalizes one of the deepest promises of ISAC: that sensing and communication are not parallel services, but mutually reinforcing parts of the same decision-making loop. The main obstacle is latency. Fast embodied systems may require sensing-to-decision loops of a few milliseconds, whereas distributed ISAC processing, edge inference, and coordination can easily exceed that budget. This makes finite-blocklength ISAC, low-latency cross-layer design, and real-time inference indispensable prerequisites. For controlled industrial and robotic environments, this direction is plausible on a relatively near horizon; for open outdoor autonomy under full environmental diversity, it remains a longer and more difficult challenge.

E. HOLOGRAPHIC RADIO

The final transformative direction pushes ISAC toward the limit of spatially continuous electromagnetic control. Holographic radio envisions very large, dense, nearly continuous apertures capable of spatial resolution approaching the diffraction limit. From a communication perspective, such apertures promise extremely fine-grained spatial multiplexing and focused energy delivery. From a sensing perspective, they imply the possibility of near-volumetric electromagnetic imaging, in which a single large surface extracts highly resolved spatial information from the environment. The joint version of this idea—holographic ISAC—would unite extremely high-capacity communication with near-diffraction-limited three-dimensional sensing.

This direction is especially important because it generalizes and extends several themes already discussed in this paper, including large-aperture arrays, near-field operation, beamfocusing, and programmable radio environments. It effectively asks what happens when the array abstraction itself is pushed toward continuity. The difficulty is that nearly every part of the system becomes more demanding at once. Element density must be extremely high, signal processing dimensionality becomes enormous, near-field channel effects dominate, and the computational burden scales rapidly with aperture size. Existing holographic MIMO theory provides only a partial foundation for these regimes, and practical holographic ISAC would require major advances in hardware manufacturing, channel modeling, and scalable inference. Still, its potential is difficult to ignore. If realized, holographic radio could redefine the boundary between communication array processing and electromagnetic imaging, making it one

of the most ambitious long-term directions in the entire ISAC roadmap.

Taken together, these five directions span a wide spectrum of maturity, risk, and potential reward. Semantic ISAC questions the informational abstraction of the field; quantum-aided ISAC questions its physical limit; neuromorphic ISAC questions its computational substrate; embodied-AI ISAC questions the separation between perception, communication, and action; and holographic radio questions the spatial abstraction of the radio interface itself. Some of these directions may remain speculative for years, while others may begin to influence system design much sooner. Their importance does not lie in whether each becomes deployable, but in the fact that each exposes a boundary of the current ISAC paradigm. In that sense, they serve as the long-horizon complement to the grand research challenges: not the problems that must be solved to complete current ISAC, but the ideas that may redefine what ISAC ultimately becomes.

IX. CONCLUSION AND OUTLOOK

ISAC has evolved from an intriguing theoretical concept into a central research imperative for next-generation wireless systems. Current standards-facing activity indicates that ISAC is becoming an important candidate capability for future 6G systems, especially through study items, use-case definition, positioning support, sensing-assisted beam management, and infrastructure-supported sensing functions. This marks a transition from purely academic exploration toward deployment-oriented evaluation, but it should not be interpreted as a hard commitment to a single Day-1 6G waveform, interface, or architecture. At the same time, hardware prototypes have begun to validate several core concepts under realistic operating conditions, demonstrating that the basic premise of joint sensing and communication is increasingly credible in practice. Yet the field remains, in several fundamental respects, young. The information-theoretic foundations for multi-user and finite-blocklength ISAC remain incomplete, ISAC-native channel models are still underdeveloped, and the privacy and security implications of pervasive networked sensing are only beginning to be addressed with the rigor they require. Since the early drafting of this roadmap, several adjacent directions have also become too important to ignore, including NTN-ISAC, low-power and backscatter-based ISAC, detection-theoretic sensing metrics, digital-twin-assisted validation, and federated sensing for privacy-preserving distributed intelligence.

A central message of this paper is that the next phase of ISAC research will be shaped by three regimes that break conventional assumptions. Near-field ISAC transforms spherical wavefront propagation from a modeling complication into a sensing opportunity, enabling range-aware beamfocusing and potentially single-shot three-dimensional localization, while simultaneously demanding new channel models and estimation frameworks. THz ISAC opens a path toward millimeter-scale resolution and extreme bandwidth, but does so under severe propagation fragility, molecular-absorption-induced band fragmentation, and wideband beam-squint effects that require joint frequency–spatial design. AI-native ISAC introduces the possibility of end-to-end learned transceivers that bypass hand-crafted modular tradeoffs, but its practical adoption is constrained by the lack of theoretical guarantees, robustness certification, and trustworthy operating principles in safety-critical settings. These regimes are not isolated. Near-field effects become especially pronounced at high frequencies because of the large aperture-to-wavelength ratio, while AI-native processing may ultimately be necessary to manage the complexity introduced by THz-wideband beam control, model mismatch, and high-dimensional sensing inference. A coherent research agenda must therefore account not only for each regime individually, but also for its interactions.

The grand research challenges identified in this paper should likewise be understood as a research agenda rather than a catalog of open questions. Together they span the full depth and breadth of the field, from foundational theory and non-asymptotic limits, to channel realism, synchronization, network coordination, privacy, security, and cross-layer integration. Their significance lies not merely in their number, but in identifying the principal points at which current ISAC progress is constrained by missing theory, models, or deployment realism. In that sense, they provide a set of waypoints for the field's near- and mid-term maturation.

Looking further ahead, the transformative directions outlined in this paper point toward a more speculative but potentially more radical future. Semantic ISAC suggests that sensing and communication may eventually be aligned at the level of task-relevant meaning rather than raw bits and low-level physical parameters, but such a direction depends on a new information theory of relevance that does not yet exist in mature form. Quantum-aided ISAC raises the possibility of approaching sensing performance closer to ultimate physical limits, yet the gap between laboratory quantum demonstrations and practical room-temperature RF wireless remains enormous. Neuromorphic radio architectures offer the promise of ultra-low-power event-driven processing, but translating continuous-valued ISAC algorithms into spiking neural computation remains highly non-trivial. ISAC for embodied AI closes the perception–communication–action loop and appears especially promising in controlled environments, though its success depends critically on progress in finite-blocklength theory, low-latency processing, and cross-layer co-design. Holographic radio pushes the field toward diffraction-limited spatial control and volumetric electromagnetic perception, but only at the price of major challenges in hardware density, near-field modeling, and tractable large-scale signal processing. These directions, therefore, differ in maturity and time horizon, but together they define the farther edge of the ISAC frontier.

The broader synthesis is that the perceptive network envisioned by ISAC—a wireless infrastructure that simultaneously connects and senses the world—is a genuinely novel

technological artifact. Its realization will require progress not only in information theory, signal processing, and hardware engineering, but also in standardization, privacy, security, machine learning, and policy-aware system design. Near-field, THz, and AI-native regimes define the present leading edge of this effort. The addition of NTN-ISAC, low-power, backscatter-based ISAC, digital twin-assisted validation, ROC-based sensing metrics, and federated sensing also shows that the ISAC roadmap is expanding beyond classical waveform and beamforming problems toward deployment- and privacy-aware, cross-domain system design. The grand research challenges organize the path forward. The transformative directions map the farther horizon. We hope this roadmap provides a useful navigational framework for the researchers, engineers, and funding agencies who will shape the ISAC-enabled future.

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